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APPLICATION NUMBER	FILING or 371(c) DATE	GRP ART UNIT	FIL FEE REC'D	ATTY. DOCKET NO	TOT CLAIMS	IND CLAIMS
63/788,821	04/15/2025		78			

206648
Spencer Southern
333 E. Bristol Rd.
Delton, MI 49046

CONFIRMATION NO. 2156
UPDATED FILING RECEIPT



0000000091150027

Date Mailed: 07/08/2025

Receipt is acknowledged of this provisional patent application. It will not be examined for patentability and will become abandoned not later than twelve months after its filing date. Any correspondence concerning the application must include the following identification information: the U.S. APPLICATION NUMBER, FILING DATE, NAME OF FIRST INVENTOR, and TITLE OF INVENTION. Fees transmitted by check or draft are subject to collection.

Please verify the accuracy of the data presented on this receipt. If an error is noted on this Filing Receipt, please submit a written request for a corrected Filing Receipt identifying the requested changes, preferably by including a properly marked-up ADS showing the changes with strike-through for deletions and underlining for additions. If you received a "Notice to File Missing Parts" or other Notice requiring a response for this application, please submit any request for correction to this Filing Receipt with your reply to the Notice. When the USPTO processes the reply to the Notice, the USPTO will generate another Filing Receipt incorporating the requested corrections provided that the request is grantable.

Inventor(s)

Spencer David Southern, Delton, MI;

Applicant(s)

Spencer David Southern, Delton, MI;

Power of Attorney: None

Permission to Access Application via Priority Document Exchange: Yes

Permission to Access Search Results: No

Applicant may provide or rescind an authorization for access using Form PTO/SB/39 or Form PTO/SB/69 as appropriate.

If Required, Foreign Filing License Granted: 05/02/2025

The country code and number of your priority application, to be used for filing abroad under the Paris Convention, is **US 63/788,821**

Projected Publication Date: None

Non-Publication Request: No

Early Publication Request: No

**** MICRO ENTITY ****

Title

Mint-to Logic System for Validation-Based Digital Routing and Protocol Execution

Statement under 37 CFR 1.55 or 1.78 for AIA (First Inventor to File) Transition Applications: No

PROTECTING YOUR INVENTION OUTSIDE THE UNITED STATES

Since the rights granted by a U.S. patent extend only throughout the territory of the United States and have no effect in a foreign country, an inventor who wishes patent protection in another country must apply for a patent in a specific country or in regional patent offices. Applicants may wish to consider the filing of an international application under the Patent Cooperation Treaty (PCT). An international (PCT) application generally has the same effect as a regular national patent application in each PCT-member country. The PCT process **simplifies** the filing of patent applications on the same invention in member countries, but **does not result** in a grant of "an international patent" and does not eliminate the need of applicants to file additional documents and fees in countries where patent protection is desired.

Almost every country has its own patent law, and a person desiring a patent in a particular country must make an application for patent in that country in accordance with its particular laws. Since the laws of many countries differ in various respects from the patent law of the United States, applicants are advised to seek guidance from specific foreign countries to ensure that patent rights are not lost prematurely.

Applicants also are advised that in the case of inventions made in the United States, the Director of the USPTO must issue a license before applicants can apply for a patent in a foreign country. The filing of a U.S. patent application serves as a request for a foreign filing license. The application's filing receipt contains further information and guidance as to the status of applicant's license for foreign filing.

Applicants may wish to consult the USPTO booklet, "General Information Concerning Patents" (specifically, the section entitled "Treaties and Foreign Patents") for more information on timeframes and deadlines for filing foreign patent applications. The guide is available either by contacting the USPTO Contact Center at 800-786-9199, or it can be viewed on the USPTO website at <http://www.uspto.gov/web/offices/pac/doc/general/index.html>.

For information on preventing theft of your intellectual property (patents, trademarks and copyrights), you may wish to consult the U.S. Government website, <http://www.stopfakes.gov>. Part of a Department of Commerce initiative, this website includes self-help "toolkits" giving innovators guidance on how to protect intellectual property in specific countries such as China, Korea and Mexico. For questions regarding patent enforcement issues, applicants may call the U.S. Government hotline at 1-866-999-HALT (1-866-999-4258).

LICENSE FOR FOREIGN FILING UNDER

Title 35, United States Code, Section 184

Title 37, Code of Federal Regulations, 5.11 & 5.15

GRANTED

The applicant has been granted a license under 35 U.S.C. 184, if the phrase "IF REQUIRED, FOREIGN FILING LICENSE GRANTED" followed by a date appears on this form. Such licenses are issued in all applications where the conditions for issuance of a license have been met, regardless of whether or not a license may be required as set forth in 37 CFR 5.15. The scope and limitations of this license are set forth in 37 CFR 5.15(a) unless an earlier

license has been issued under 37 CFR 5.15(b). The license is subject to revocation upon written notification. The date indicated is the effective date of the license, unless an earlier license of similar scope has been granted under 37 CFR 5.13 or 5.14.

This license is to be retained by the licensee and may be used at any time on or after the effective date thereof unless it is revoked. This license is automatically transferred to any related applications(s) filed under 37 CFR 1.53(d). This license is not retroactive.

The grant of a license does not in any way lessen the responsibility of a licensee for the security of the subject matter as imposed by any Government contract or the provisions of existing laws relating to espionage and the national security or the export of technical data. Licensees should apprise themselves of current regulations especially with respect to certain countries, of other agencies, particularly the Office of Defense Trade Controls, Department of State (with respect to Arms, Munitions and Implements of War (22 CFR 121-128)); the Bureau of Industry and Security, Department of Commerce (15 CFR parts 730-774); the Office of Foreign Assets Control, Department of Treasury (31 CFR Parts 500+) and the Department of Energy.

NOT GRANTED

No license under 35 U.S.C. 184 has been granted at this time, if the phrase "IF REQUIRED, FOREIGN FILING LICENSE GRANTED" DOES NOT appear on this form. Applicant may still petition for a license under 37 CFR 5.12, if a license is desired before the expiration of 6 months from the filing date of the application. If 6 months has lapsed from the filing date of this application and the licensee has not received any indication of a secrecy order under 35 U.S.C. 181, the licensee may foreign file the application pursuant to 37 CFR 5.15(b).

SelectUSA

The United States represents the largest, most dynamic marketplace in the world and is an unparalleled location for business investment, innovation, and commercialization of new technologies. The U.S. offers tremendous resources and advantages for those who invest and manufacture goods here. Through SelectUSA, our nation works to promote and facilitate business investment. SelectUSA provides information assistance to the international investor community; serves as an ombudsman for existing and potential investors; advocates on behalf of U.S. cities, states, and regions competing for global investment; and counsels U.S. economic development organizations on investment attraction best practices. To learn more about why the United States is the best country in the world to develop technology, manufacture products, deliver services, and grow your business, visit <http://www.SelectUSA.gov> or call +1-202-482-6800.

TO: spencersouthern12@gmail.com
FROM: noreply@uspto.gov
CC: patentcenter_eofficeaction@uspto.gov
SUBJECT: USPTO: Patent Electronic System - Correspondence Notification for Customer Number 206648

Tue Jul 08 05:01:38 EDT 2025

Dear Patent Center Customer:

Correspondence Address:

Spencer Southern
333 E. Bristol Rd.
Delton, MICHIGAN, 49046
UNITED STATES

This is a courtesy notification regarding the following USPTO patent application(s) associated with your Customer Number, 206648, that have new outgoing correspondence. This correspondence is now available for viewing in Patent Center.

The official date of notification of the outgoing correspondence will be indicated on the form (e.g., PTOL-90) accompanying the correspondence.

Disclaimer:

The list of documents shown below are provided as a courtesy and is not part of the official file wrapper. The content of the images shown in the Image File Wrapper is the official record.

Application	Document	Mailroom Date	Attorney Docket No.
63/88821	APP.FILE.REC	07/08/2025	

To view your correspondence online, please sign in to [Patent Center](#) and then select Workbench/View correspondence. To update your email address(es), select Manage/Manage customer numbers.

If you have any questions, please contact the [Patent Electronic Business Center](#) (EBC) at ebc@uspto.gov or 866-217-9197 Monday – Friday, 6 a.m. to midnight ET.

Please do not reply to this email as it was sent from an unmonitored mailbox.

Sincerely,

The Patent Center Team



United States Patent and Trademark Office

Office of the Chief Financial Officer

Document Code:WFEE

User :Almaz Sahle

Sale Accounting Date:07/07/2025

Sale Item Reference Number
63788821

Effective Date
07/02/2025

Document Number	Fee Code	Fee Code Description	Amount Paid	Payment Method
I202577A55207218	3005	PROVISIONAL APPLICATION FILING FEE	\$65.00	Salea

Acknowledgement of Change to Micro Entity Status

APPLICATION #	FILING OR 371(C) DATE	FIRST NAMED APPLICANT	ATTORNEY DOCKET #	REQUEST ID
63/788,821	04/15/2025	Spencer David Southern		430920

The entity status change request below filed through Patent Center on 07/05/2025 has been accepted.

Certifications

CERTIFICATION OF MICRO ENTITY STATUS (GROSS INCOME BASIS)

The applicant hereby certifies the following:

- (1) SMALL ENTITY REQUIREMENT -The applicant qualifies as a small entity as defined in 37 CFR 1.27.
- (2) APPLICATION FILING LIMIT - Neither the applicant nor the inventor nor a joint inventor has been named as the inventor or a joint inventor on more than four previously filed U.S. patent applications, excluding provisional applications and international applications under the Patent Cooperation Treaty (PCT) for which the basic national fee under 37 CFR 1.492(a) was not paid, and also excluding patent applications for which the applicant has assigned all ownership rights, or is obligated to assign all ownership rights, as a result of the applicant's previous employment.
- (3) GROSS INCOME LIMIT ON APPLICANTS AND INVENTORS - Neither the applicant nor the inventor nor a joint inventor, in the calendar year preceding the calendar year in which the applicable fee is being paid, had a gross income, as defined in section 61(a) of the Internal Revenue Code of 1986 (26 U.S.C. 61(a)), exceeding the Maximum Qualifying Gross Income reported on the USPTO website at http://www.uspto.gov/patents/law/micro_entity.jsp which is equal to three times the median household income for that preceding calendar year, as most recently reported by the Bureau of the Census.
- (4) GROSS INCOME LIMIT ON PARTIES WITH AN OWNERSHIP INTEREST - Neither the applicant nor the inventor nor a joint inventor has assigned, granted, or conveyed, nor is under an obligation by contract or law to assign, grant, or convey, a license or other ownership interest in the application concerned to an entity that, in the calendar year preceding the calendar year in which the applicable fee is being paid, had a gross income, as defined in section 61(a) of the Internal Revenue Code of 1986, exceeding the Maximum Qualifying Gross Income reported on the USPTO website at http://www.uspto.gov/patents/law/micro_entity.jsp which is equal to three times the median household income for that preceding calendar year, as most recently reported by the Bureau of the Census.

Signature

I certify, in accordance with 37 CFR 1.4(d)(4), that I am one of the signatories making the entity status change.

Signature

Name

Registration #

/Spencer Southern/

SPENCER SOUTHERN



UNITED STATES
PATENT AND TRADEMARK OFFICE

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Alexandria, VA 22313 - 1450
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ELECTRONIC PAYMENT RECEIPT

APPLICATION #	RECEIPT DATE / TIME	ATTORNEY DOCKET #
63/788,821	07/02/2025 06:46:11 PM Z ET	-

Title of Invention

Mint-to Logic System for Validation-Based Digital Routing and Protocol Execution

Application Information

APPLICATION TYPE	Utility - Provisional Application under 35 USC 111(b)	PATENT #	-
CONFIRMATION #	2156	FILED BY	SPENCER SOUTHERN
PATENT CENTER #	71197581	AUTHORIZED BY	-
CUSTOMER #	206648	FILING DATE	04/15/2025
CORRESPONDENCE ADDRESS	-	FIRST NAMED INVENTOR	Spencer David Southern

Payment Information

PAYMENT METHOD	PAYMENT TRANSACTION ID	PAYMENT AUTHORIZED BY
CARD / 0279	E202572I48508102	SPENCER SOUTHERN

FEE CODE	DESCRIPTION	ITEM PRICE(\$)	QUANTITY	ITEM TOTAL(\$)
3052	SURCHARGE - LATE PROVISIONAL FILING FEE OR COVER SHEET	13.00	1	13.00
TOTAL AMOUNT:				\$13.00

This Acknowledgement Receipt evidences receipt on the noted date by the USPTO of the indicated documents, characterized by the applicant, and including page counts, where applicable. It serves as evidence of receipt similar to a Post Card, as described in MPEP 503.

New Applications Under 35 U.S.C. 111

If a new application is being filed and the application includes the necessary components for filing date (see 37 CFR 1.53(b)-(d) and MPEP 506), a Filing Receipt (37 CFR 1.54) will be issued in due course and the date shown on this Acknowledgement Receipt will establish the filing date of the application

National Stage of an International Application under 35 U.S.C. 371

If a timely submission to enter the national stage of an international application is compliant with the conditions of 35 U.S.C.

371 and other applicable requirements a Form PCT/DO/EO/903 indicating acceptance of the application as a national stage submission under 35 U.S.C. 371 will be issued in addition to the Filing Receipt, in due course.

New International Application Filed with the USPTO as a Receiving Office

If a new international application is being filed and the international application includes the necessary components for an international filing date (see PCT Article 11 and MPEP 1810), a Notification of the International Application Number and of the International Filing Date (Form PCT/RO/105) will be issued in due course, subject to prescriptions concerning national security, and the date shown on this Acknowledgement Receipt will establish the international filing date of the application.



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PATENT AND TRADEMARK OFFICE

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ELECTRONIC ACKNOWLEDGEMENT RECEIPT

APPLICATION #	RECEIPT DATE / TIME	ATTORNEY DOCKET #
63/788,821	07/02/2025 06:46:11 PM Z ET	-

Title of Invention

Mint-to Logic System for Validation-Based Digital Routing and Protocol Execution

Application Information

APPLICATION TYPE	Utility - Provisional Application under 35 USC 111(b)	PATENT #	-
CONFIRMATION #	2156	FILED BY	SPENCER SOUTHERN
PATENT CENTER #	71197581	FILING DATE	04/15/2025
CUSTOMER #	206648	FIRST NAMED INVENTOR	Spencer David Southern
CORRESPONDENCE ADDRESS	-	AUTHORIZED BY	-

Documents

TOTAL DOCUMENTS: 0

DOCUMENT	PAGES	DESCRIPTION	SIZE (KB)
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Digest

DOCUMENT	MESSAGE DIGEST(SHA-512)
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If a timely submission to enter the national stage of an international application is compliant with the conditions of 35 U.S.C. 371 and other applicable requirements a Form PCT/DO/EO/903 indicating acceptance of the application as a national stage

submission under 35 U.S.C. 371 will be issued in addition to the Filing Receipt, in due course.

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UNITED STATES
PATENT AND TRADEMARK OFFICEP.O. Box 1450
Alexandria, VA 22313 - 1450
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ELECTRONIC ACKNOWLEDGEMENT RECEIPT

APPLICATION #
63/788,821RECEIPT DATE / TIME
07/02/2025 10:45:02 AM Z ETATTORNEY DOCKET #
-

Title of Invention

Mint-to Logic System for Validation-Based Digital Routing and Protocol Execution

Application Information

APPLICATION TYPE Utility - Provisional Application under
35 USC 111(b)

PATENT # -

CONFIRMATION # 2156

FILED BY SPENCER SOUTHERN

PATENT CENTER # 71184778

FILING DATE 04/15/2025

CUSTOMER # 206648

FIRST NAMED
INVENTOR Spencer David SouthernCORRESPONDENCE
ADDRESS -

AUTHORIZED BY -

Documents

TOTAL DOCUMENTS: 2

DOCUMENT	PAGES	DESCRIPTION	SIZE (KB)
back sheet micro entity HPSCAN_202507021437464 15_2025-07- 02_143833721.pdf	1	Certification of Micro Entity (Gross Income Basis)	1144 KB
micro entity HPSCAN_202507021435277 33_2025-07- 02_143614911.pdf	1	Certification of Micro Entity (Gross Income Basis)	1263 KB

Digest

DOCUMENT	MESSAGE DIGEST(SHA-512)
back sheet micro entity	1AFD76CCFC4FAED20B0441876E68A4B18B8033BBF21C9ECB

HPSCAN_20250702143746415	6098A096FE19DF2F6A9B038928EC6DBC6E07EE29F753D7103
_2025-07-02_143833721.pdf	8C59944C2F087607363563F2AF230BA
micro entity	BADC7B987903CBEDBDF7BE43366CB300883A3581240F54291
HPSCAN_20250702143527733	98B1AD14E8C34267309190BE7814046CB318662E3D9AD4B69
_2025-07-02_143614911.pdf	BE3FB2749E70C3BE089947C935AD8B

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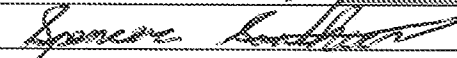
**CERTIFICATION OF MICRO ENTITY STATUS
(GROSS INCOME BASIS)**

Application Number or Control Number (if applicable): 63788821	Patent Number (if applicable):
First Named Inventor: Spencer D. Southern	Title of Invention: Mint-to Logic System for Validation-Based Digital Routing and Protocol Execution

The applicant hereby certifies the following—

- (1) **SMALL ENTITY REQUIREMENT** – The applicant qualifies as a small entity as defined in 37 CFR 1.27.
- (2) **APPLICATION FILING LIMIT** – Neither the applicant nor the inventor nor a joint inventor has been named as the inventor or a joint inventor on more than four previously filed U.S. patent applications, excluding provisional applications and international applications under the Patent Cooperation Treaty (PCT) for which the basic national fee under 37 CFR 1.492(a) was not paid, and also excluding patent applications for which the applicant has assigned all ownership rights, or is obligated to assign all ownership rights, as a result of the applicant's previous employment.
- (3) **GROSS INCOME LIMIT ON APPLICANTS AND INVENTORS** – Neither the applicant nor the inventor nor a joint inventor, in the calendar year preceding the calendar year in which the applicable fee is being paid, had a gross income, as defined in section 61(a) of the Internal Revenue Code of 1986 (26 U.S.C. 61(a)), exceeding the "Maximum Qualifying Gross Income" reported on the USPTO Web site at http://www.uspto.gov/patents/law/micro_entity.jsp which is equal to three times the median household income for that preceding calendar year, as most recently reported by the Bureau of the Census.
- (4) **GROSS INCOME LIMIT ON PARTIES WITH AN "OWNERSHIP INTEREST"** – Neither the applicant nor the inventor nor a joint inventor has assigned, granted, or conveyed, nor is under an obligation by contract or law to assign, grant, or convey, a license or other ownership interest in the application concerned to an entity that, in the calendar year preceding the calendar year in which the applicable fee is being paid, had a gross income, as defined in section 61(a) of the Internal Revenue Code of 1986, exceeding the "Maximum Qualifying Gross Income" reported on the USPTO Web site at http://www.uspto.gov/patents/law/micro_entity.jsp which is equal to three times the median household income for that preceding calendar year, as most recently reported by the Bureau of the Census.

SIGNATURE by an authorized party set forth in 37 CFR 1.33(b)

Signature				
Name	Spencer D. Southern			
Date	07/02/2025	Telephone	2693318435	Registration No.
☐	There is more than one inventor and I am one of the inventors who are jointly identified as the applicant. The required additional certification form(s) signed by the other joint inventor(s) are included with this form.			

Privacy Act Statement

The Privacy Act of 1974 (P.L. 93-579) requires that you be given certain information in connection with your submission of the attached form related to a patent application or patent. The United States Patent and Trademark Office (USPTO) collects the information in this record under authority of 35 U.S.C. 2. The USPTO's system of records is used to manage all applicant and owner information including name, citizenship, residence, post office address, and other information with respect to inventors and their legal representatives pertaining to the applicant's/owner's activities in connection with the invention for which a patent is sought or has been granted. The applicable Privacy Act System of Records Notice for the information collected in this form is COMMERCE/PAT-TM-7 Patent Application Files, available in the Federal Register at 78 FR 19243 (March 29, 2013). <https://www.govinfo.gov/content/pkg/FR-2013-03-29/pdf/2013-07341.pdf>

Routine uses of the information in this record may include disclosure to: 1) law enforcement, in the event that the system of records indicates a violation or potential violation of law; 2) a Federal, state, local, or international agency, in response to its request; 3) a contractor of the USPTO having need for the information in order to perform a contract; 4) the Department of Justice for determination of whether the Freedom of Information Act (FOIA) requires disclosure of the record; 5) a Member of Congress submitting a request involving an individual to whom the record pertains, when the individual has requested the Member's assistance with respect to the subject matter of the record; 6) a court, magistrate, or administrative tribunal, in the course of presenting evidence, including disclosures to opposing counsel in the course of settlement negotiations; 7) the Administrator, General Services Administration (GSA), or their designee, during an inspection of records conducted by GSA under authority of 44 U.S.C. 2904 and 2906, in accordance with the GSA regulations and any other relevant (i.e., GSA or Commerce) directive, where such disclosure shall not be used to make determinations about individuals; 8) another federal agency for purposes of National Security review (35 U.S.C. 181) and for review pursuant to the Atomic Energy Act (42 U.S.C. 218(c)); 9) the Office of Personnel Management (OPM) for personnel research purposes; and 9) the Office of Management and Budget (OMB) for legislative coordination and clearance.

If you do not furnish the information requested on this form, the USPTO may not be able to process and/or examine your submission, which may result in termination of proceedings, abandonment of the application, and/or expiration of the patent.

Additional Uses

Additional USPTO uses of the information in this record may include disclosure to: 1) the International Bureau of the World Intellectual Property Organization, if the record is related to an international application filed under the Patent Cooperation Treaty; 2) the public i) after publication of the application pursuant to 35 U.S.C. 122(b), ii) after issuance of a patent pursuant to 35 U.S.C. 151, iii) if the record was filed in an application which became abandoned or in which the proceedings were terminated and which application is referenced by either a published application, an application open to public inspections, or an issued patent, or iv) without publication of the application or patent under the specific circumstances provided for by 37 CFR 1.14(a)(1)(v)-(vii); and/or 3) the National Archives and Records Administration, for inspection of records.



UNITED STATES PATENT AND TRADEMARK OFFICE

UNITED STATES DEPARTMENT OF COMMERCE
United States Patent and Trademark Office
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APPLICATION NUMBER	FILING OR 371(C) DATE	FIRST NAMED APPLICANT	ATTY. DOCKET NO./TITLE
63/788,821	04/15/2025	Spencer David Southern	

206648
Spencer Southern
333 E. Bristol Rd.
Delton, MI 49046

CONFIRMATION NO. 2156

FORMALITIES LETTER



OC00000090954246

Date Mailed: 07/02/2025

NOTICE OF INCOMPLETE REPLY (PROVISIONAL)

Filing Date Granted

The U.S. Patent and Trademark Office received your reply on 06/29/2025 to the NOTICE TO FILE MISSING PARTS OF PROVISIONAL APPLICATION (Notice) mailed 05/06/2025 and it has been entered into the application. The reply, however, is not a complete reply because all of the items required by the NOTICE TO FILE MISSING PARTS OF PROVISIONAL APPLICATION have not been received. See "Items Required to Avoid Abandonment" section below. A complete reply must be timely filed to prevent ABANDONMENT of the above-identified application.

The period for reply continues to run from the date of the NOTICE TO FILE MISSING PARTS OF PROVISIONAL APPLICATION mailed 05/06/2025. Applicant must submit all required items and pay any fees required below within two months from the date of the NOTICE TO FILE MISSING PARTS OF PROVISIONAL APPLICATION to avoid abandonment. If the reply is submitted after two months from the date of the NOTICE TO FILE MISSING PARTS OF PROVISIONAL APPLICATION, extensions of time may be obtained by filing a petition accompanied by the extension fee under the provisions of 37 CFR 1.136(a). See "Time Period for Reply" section below.

Items Required to Avoid Abandonment:

The required items noted below SHOULD be filed along with any items required above.

- The statutory basic filing fee is insufficient based on the current entity status of the application. The remainder of the basic filing fee or a proper establishment of a discounted entity status must be submitted.
- Surcharge as set forth in 37 CFR 1.16(g) has not been received.

The surcharge is due for late submission of filing fee or cover sheet.

SUMMARY OF FEES DUE:

The fee(s) required (including the fees noted above) are as follows. Small entity discount is in effect. If applicant is qualified for micro entity status, an acceptable Certification of Micro Entity Status must be submitted to establish micro entity status. (See 37 CFR 1.29 and forms PTO/SB/15A and 15B.)

- **\$130** basic filing fee.
- **\$26** surcharge.
- **\$(65)** previous unapplied payment amount.
- **\$91** TOTAL FEE BALANCE DUE.

Applicant is notified that the above-identified application contains the deficiencies noted below. No period for reply is set forth in this Notice for correction of these deficiencies. However, if a deficiency relates to the inventor's oath or declaration, the applicant must file an oath or declaration in compliance with 37 CFR 1.63, or a substitute statement in compliance with 37 CFR 1.64, executed by or with respect to each actual inventor no later than the expiration of the time period set in the "Notice of Allowability" to avoid abandonment. See 37 CFR 1.53(f).

Items Required To Avoid Processing Delays:

The deficiencies indicated below should be addressed promptly to avoid further processing delays.

This application, which was filed with an indication of micro entity status, fails to meet the requirements for establishing micro entity status because:

- The certification of micro entity status is unsigned. A certification of micro entity status must be signed by an authorized party under 37 CFR 1.33(b). (Authorized parties include a registered attorney or agent, the sole inventor identified as the applicant, or all the joint inventors identified as the applicant except that one joint inventor applicant may sign if appointed to prosecute the application by all the other joint inventors.)

If micro entity status is established in reply to this notice by filing an acceptable Certification of Micro Entity Status, the fees itemized in the SUMMARY OF FEES DUE section of this Notice will be reduced to reflect the appropriate micro entity discount, and any previous payment amount may be applied to the discounted fees.

If applicant is not entitled to micro entity status under 37 CFR 1.29, applicant may still qualify for an appropriate reduction of fees if small entity status can be established under 37 CFR 1.27.

Time Period for Reply:

Mail date of Notice: 05/06/2025

Last date that extension may be obtained: (Note: The petition and fee must be received 12/08/2025 by this date, or include a proper certificate of mailing under 37 CFR 1.8 with a date on or before this date, and extend the time to include this date.)

The following table sets forth the possible extensions of time available, depending on when the reply and corresponding extension of time fee are received.

Length of Extension of Time (response due on or before)	Fee under 37 CFR 1.17(a) effective March 19, 2013		
	Undiscounted	Small Entity	Micro Entity
One Month (08/06/2025)	\$50	\$20	\$10
Two Months (09/08/2025)	\$100	\$40	\$20
Three Months (10/06/2025)	\$200	\$80	\$40
Four Months (11/06/2025)	\$400	\$160	\$80
Five Months (12/08/2025)	\$800	\$320	\$160

Replies must be received in the USPTO within the set time period or must include a proper Certificate of Mailing or Transmission under 37 CFR 1.8 with a mailing or transmission date within the set time period. For more information and a suggested format, see Form PTO/SB/92 and MPEP 512.

Replies should be addressed to:

Mail Stop Missing Parts
Commissioner for Patents
P.O. Box 1450
Alexandria VA 22313-1450

REPLY VIA ELECTRONIC FILING

A reply may be submitted via electronic filing only by registered users of the appropriate USPTO patent electronic filing system and must include:

- A separate document having the document description, "**Applicant response to Pre-Exam Formalities Notice**"; and
- the item(s) for which a reply period is set forth above.

For more information about electronic filing, contact the Patent Electronic Business Center (EBC) 1-866-217-9197 or visit <https://www.uspto.gov/ebc>

NOTE: If the reply is not being submitted electronically, a copy of this notice must be included.

Questions about the contents of this notice and the requirements it sets forth should be directed to the Office of Data Management, Application Assistance Unit, at **(571) 272-4000** or **(571) 272-4200** or **1-888-786-0101**.

/byemane-brehan/

TO: spencersouthern12@gmail.com
FROM: noreply@uspto.gov
CC: patentcenter_eofficeaction@uspto.gov
SUBJECT: USPTO: Patent Electronic System - Correspondence Notification for Customer Number 206648

Wed Jul 02 05:02:13 EDT 2025

Dear Patent Center Customer:

Correspondence Address:

Spencer Southern
333 E. Bristol Rd.
Delton, MICHIGAN, 49046
UNITED STATES

This is a courtesy notification regarding the following USPTO patent application(s) associated with your Customer Number, 206648, that have new outgoing correspondence. This correspondence is now available for viewing in Patent Center.

The official date of notification of the outgoing correspondence will be indicated on the form (e.g., PTOL-90) accompanying the correspondence.

Disclaimer:

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Application	Document	Mailroom Date	Attorney Docket No.
63/88821	NT.INC.REPLY	07/02/2025	

To view your correspondence online, please sign in to [Patent Center](#) and then select Workbench/View correspondence. To update your email address(es), select Manage/Manage customer numbers.

If you have any questions, please contact the [Patent Electronic Business Center](#) (EBC) at ebc@uspto.gov or 866-217-9197 Monday – Friday, 6 a.m. to midnight ET.

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Sincerely,

The Patent Center Team



UNITED STATES
PATENT AND TRADEMARK OFFICE

P.O. Box 1450
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ELECTRONIC ACKNOWLEDGEMENT RECEIPT

APPLICATION #
63/788,821

RECEIPT DATE / TIME
06/29/2025 08:34:05 PM Z ET

ATTORNEY DOCKET #
-

Title of Invention

Mint-to Logic System for Validation-Based Digital Routing and Protocol Execution

Application Information

APPLICATION TYPE Utility - Provisional Application under
35 USC 111(b)

PATENT # -

CONFIRMATION # 2156

FILED BY SPENCER SOUTHERN

PATENT CENTER # 71134728

FILING DATE 04/15/2025

CUSTOMER # 206648

FIRST NAMED INVENTOR Spencer David Southern

CORRESPONDENCE ADDRESS -

AUTHORIZED BY -

Documents

TOTAL DOCUMENTS: 1

DOCUMENT	PAGES	DESCRIPTION	SIZE (KB)
sb0015a.pdf	2	Certification of Micro Entity (Gross Income Basis)	188 KB

Digest

DOCUMENT	MESSAGE DIGEST(SHA-512)
sb0015a.pdf	0468AFC7353A6C721D0F9E50757FBBF4AB752439E68222E5140 BFCBB2FC132E5782B5C52846E3C59BB8B5DFB2EABF698EC2 E26FF02B8C4C20C24E4B0540FA0AF1

This Acknowledgement Receipt evidences receipt on the noted date by the USPTO of the indicated documents, characterized by the applicant, and including page counts, where applicable. It serves as evidence of receipt similar to a Post Card, as described in MPEP 503.

New Applications Under 35 U.S.C. 111

If a new application is being filed and the application includes the necessary components for filing date (see 37 CFR 1.53(b)-(d) and MPEP 506), a Filing Receipt (37 CFR 1.54) will be issued in due course and the date shown on this Acknowledgement Receipt will establish the filing date of the application

National Stage of an International Application under 35 U.S.C. 371

If a timely submission to enter the national stage of an international application is compliant with the conditions of 35 U.S.C. 371 and other applicable requirements a Form PCT/DO/EO/903 indicating acceptance of the application as a national stage submission under 35 U.S.C. 371 will be issued in addition to the Filing Receipt, in due course.

New International Application Filed with the USPTO as a Receiving Office

If a new international application is being filed and the international application includes the necessary components for an international filing date (see PCT Article 11 and MPEP 1810), a Notification of the International Application Number and of the International Filing Date (Form PCT/RO/105) will be issued in due course, subject to prescriptions concerning national security, and the date shown on this Acknowledgement Receipt will establish the international filing date of the application.

CERTIFICATION OF MICRO ENTITY STATUS (GROSS INCOME BASIS)

Application Number or Control Number (if applicable): 63788821	Patent Number (if applicable):
First Named Inventor: Spencer D. Southern	Title of Invention: Mint-to Logic System for Validation-Based Digital Routing and Protocol Execution

The applicant hereby certifies the following—

- (1) **SMALL ENTITY REQUIREMENT** – The applicant qualifies as a small entity as defined in 37 CFR 1.27.

- (2) **APPLICATION FILING LIMIT** – Neither the applicant nor the inventor nor a joint inventor has been named as the inventor or a joint inventor on more than four previously filed U.S. patent applications, excluding provisional applications and international applications under the Patent Cooperation Treaty (PCT) for which the basic national fee under 37 CFR 1.492(a) was not paid, and also excluding patent applications for which the applicant has assigned all ownership rights, or is obligated to assign all ownership rights, as a result of the applicant's previous employment.

- (3) **GROSS INCOME LIMIT ON APPLICANTS AND INVENTORS** – Neither the applicant nor the inventor nor a joint inventor, in the calendar year preceding the calendar year in which the applicable fee is being paid, had a gross income, as defined in section 61(a) of the Internal Revenue Code of 1986 (26 U.S.C. 61(a)), exceeding the "Maximum Qualifying Gross Income" reported on the USPTO Web site at http://www.uspto.gov/patents/law/micro_entity.jsp which is equal to three times the median household income for that preceding calendar year, as most recently reported by the Bureau of the Census.

- (4) **GROSS INCOME LIMIT ON PARTIES WITH AN "OWNERSHIP INTEREST"** – Neither the applicant nor the inventor nor a joint inventor has assigned, granted, or conveyed, nor is under an obligation by contract or law to assign, grant, or convey, a license or other ownership interest in the application concerned to an entity that, in the calendar year preceding the calendar year in which the applicable fee is being paid, had a gross income, as defined in section 61(a) of the Internal Revenue Code of 1986, exceeding the "Maximum Qualifying Gross Income" reported on the USPTO Web site at http://www.uspto.gov/patents/law/micro_entity.jsp which is equal to three times the median household income for that preceding calendar year, as most recently reported by the Bureau of the Census.

SIGNATURE by an authorized party set forth in 37 CFR 1.33(b)

Signature	Spencer D. Southern				
Name	Spencer D. Southern				
Date	06/29/2025	Telephone	2693318435	Registration No.	
☐	There is more than one inventor and I am one of the inventors who are jointly identified as the applicant. The required additional certification form(s) signed by the other joint inventor(s) are included with this form.				

Privacy Act Statement

The Privacy Act of 1974 (P.L. 93-579) requires that you be given certain information in connection with your submission of the attached form related to a patent application or patent. The United States Patent and Trademark Office (USPTO) collects the information in this record under authority of 35 U.S.C. 2. The USPTO's system of records is used to manage all applicant and owner information including name, citizenship, residence, post office address, and other information with respect to inventors and their legal representatives pertaining to the applicant's/owner's activities in connection with the invention for which a patent is sought or has been granted. The applicable Privacy Act System of Records Notice for the information collected in this form is COMMERCE/PAT-TM-7 Patent Application Files, available in the Federal Register at 78 FR 19243 (March 29, 2013). <https://www.govinfo.gov/content/pkg/FR-2013-03-29/pdf/2013-07341.pdf>

Routine uses of the information in this record may include disclosure to: 1) law enforcement, in the event that the system of records indicates a violation or potential violation of law; 2) a Federal, state, local, or international agency, in response to its request; 3) a contractor of the USPTO having need for the information in order to perform a contract; 4) the Department of Justice for determination of whether the Freedom of Information Act (FOIA) requires disclosure of the record; 5) a Member of Congress submitting a request involving an individual to whom the record pertains, when the individual has requested the Member's assistance with respect to the subject matter of the record; 6) a court, magistrate, or administrative tribunal, in the course of presenting evidence, including disclosures to opposing counsel in the course of settlement negotiations; 7) the Administrator, General Services Administration (GSA), or their designee, during an inspection of records conducted by GSA under authority of 44 U.S.C. 2904 and 2906, in accordance with the GSA regulations and any other relevant (i.e., GSA or Commerce) directive, where such disclosure shall not be used to make determinations about individuals; 8) another federal agency for purposes of National Security review (35 U.S.C. 181) and for review pursuant to the Atomic Energy Act (42 U.S.C. 218(c)); 9) the Office of Personnel Management (OPM) for personnel research purposes; and 9) the Office of Management and Budget (OMB) for legislative coordination and clearance.

If you do not furnish the information requested on this form, the USPTO may not be able to process and/or examine your submission, which may result in termination of proceedings, abandonment of the application, and/or expiration of the patent.

Additional Uses

Additional USPTO uses of the information in this record may include disclosure to: 1) the International Bureau of the World Intellectual Property Organization, if the record is related to an international application filed under the Patent Cooperation Treaty; 2) the public i) after publication of the application pursuant to 35 U.S.C. 122(b), ii) after issuance of a patent pursuant to 35 U.S.C. 151, iii) if the record was filed in an application which became abandoned or in which the proceedings were terminated and which application is referenced by either a published application, an application open to public inspections, or an issued patent, or iv) without publication of the application or patent under the specific circumstances provided for by 37 CFR 1.14(a)(1)(v)-(vii); and/or 3) the National Archives and Records Administration, for inspection of records.



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United States Patent and Trademark Office
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Alexandria, Virginia 22313-1450
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APPLICATION NUMBER	FILING or 371(c) DATE	GRP ART UNIT	FIL FEE REC'D	ATTY. DOCKET NO.	TOT CLAIMS	IND CLAIMS
63/788,821	04/15/2025		65			

206648
Spencer Southern
333 E. Bristol Rd.
Delton, MI 49046

CONFIRMATION NO. 2156
FILING RECEIPT



0000000088120628

Date Mailed: 05/06/2025

Receipt is acknowledged of this provisional patent application. It will not be examined for patentability and will become abandoned not later than twelve months after its filing date. Any correspondence concerning the application must include the following identification information: the U.S. APPLICATION NUMBER, FILING DATE, NAME OF FIRST INVENTOR, and TITLE OF INVENTION. Fees transmitted by check or draft are subject to collection.

Please verify the accuracy of the data presented on this receipt. If an error is noted on this Filing Receipt, please submit a written request for a corrected Filing Receipt identifying the requested changes, preferably by including a properly marked-up ADS showing the changes with strike-through for deletions and underlining for additions. If you received a "Notice to File Missing Parts" or other Notice requiring a response for this application, please submit any request for correction to this Filing Receipt with your reply to the Notice. When the USPTO processes the reply to the Notice, the USPTO will generate another Filing Receipt incorporating the requested corrections provided that the request is grantable.

Inventor(s)

Spencer David Southern, Delton, MI;

Applicant(s)

Spencer David Southern, Delton, MI;

Power of Attorney: None

Permission to Access Application via Priority Document Exchange: Yes

Permission to Access Search Results: No

Applicant may provide or rescind an authorization for access using Form PTO/SB/39 or Form PTO/SB/69 as appropriate.

If Required, Foreign Filing License Granted: 05/02/2025

The country code and number of your priority application, to be used for filing abroad under the Paris Convention, is **US 63/788,821**

Projected Publication Date: None

Non-Publication Request: No

Early Publication Request: No

**** SMALL ENTITY ****

Title

Mint-to Logic System for Validation-Based Digital Routing and Protocol Execution

Statement under 37 CFR 1.55 or 1.78 for AIA (First Inventor to File) Transition Applications: No

PROTECTING YOUR INVENTION OUTSIDE THE UNITED STATES

Since the rights granted by a U.S. patent extend only throughout the territory of the United States and have no effect in a foreign country, an inventor who wishes patent protection in another country must apply for a patent in a specific country or in regional patent offices. Applicants may wish to consider the filing of an international application under the Patent Cooperation Treaty (PCT). An international (PCT) application generally has the same effect as a regular national patent application in each PCT-member country. The PCT process **simplifies** the filing of patent applications on the same invention in member countries, but **does not result** in a grant of "an international patent" and does not eliminate the need of applicants to file additional documents and fees in countries where patent protection is desired.

Almost every country has its own patent law, and a person desiring a patent in a particular country must make an application for patent in that country in accordance with its particular laws. Since the laws of many countries differ in various respects from the patent law of the United States, applicants are advised to seek guidance from specific foreign countries to ensure that patent rights are not lost prematurely.

Applicants also are advised that in the case of inventions made in the United States, the Director of the USPTO must issue a license before applicants can apply for a patent in a foreign country. The filing of a U.S. patent application serves as a request for a foreign filing license. The application's filing receipt contains further information and guidance as to the status of applicant's license for foreign filing.

Applicants may wish to consult the USPTO booklet, "General Information Concerning Patents" (specifically, the section entitled "Treaties and Foreign Patents") for more information on timeframes and deadlines for filing foreign patent applications. The guide is available either by contacting the USPTO Contact Center at 800-786-9199, or it can be viewed on the USPTO website at <http://www.uspto.gov/web/offices/pac/doc/general/index.html>.

For information on preventing theft of your intellectual property (patents, trademarks and copyrights), you may wish to consult the U.S. Government website, <http://www.stopfakes.gov>. Part of a Department of Commerce initiative, this website includes self-help "toolkits" giving innovators guidance on how to protect intellectual property in specific countries such as China, Korea and Mexico. For questions regarding patent enforcement issues, applicants may call the U.S. Government hotline at 1-866-999-HALT (1-866-999-4258).

LICENSE FOR FOREIGN FILING UNDER

Title 35, United States Code, Section 184

Title 37, Code of Federal Regulations, 5.11 & 5.15

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This license is to be retained by the licensee and may be used at any time on or after the effective date thereof unless it is revoked. This license is automatically transferred to any related applications(s) filed under 37 CFR 1.53(d). This license is not retroactive.

The grant of a license does not in any way lessen the responsibility of a licensee for the security of the subject matter as imposed by any Government contract or the provisions of existing laws relating to espionage and the national security or the export of technical data. Licensees should apprise themselves of current regulations especially with respect to certain countries, of other agencies, particularly the Office of Defense Trade Controls, Department of State (with respect to Arms, Munitions and Implements of War (22 CFR 121-128)); the Bureau of Industry and Security, Department of Commerce (15 CFR parts 730-774); the Office of Foreign Assets Control, Department of Treasury (31 CFR Parts 500+) and the Department of Energy.

NOT GRANTED

No license under 35 U.S.C. 184 has been granted at this time, if the phrase "IF REQUIRED, FOREIGN FILING LICENSE GRANTED" DOES NOT appear on this form. Applicant may still petition for a license under 37 CFR 5.12, if a license is desired before the expiration of 6 months from the filing date of the application. If 6 months has lapsed from the filing date of this application and the licensee has not received any indication of a secrecy order under 35 U.S.C. 181, the licensee may foreign file the application pursuant to 37 CFR 5.15(b).

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Alexandria, Virginia 22313-1450
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APPLICATION NUMBER	FILING OR 371(C) DATE	FIRST NAMED APPLICANT	ATTY. DOCKET NO./TITLE
63/788,821	04/15/2025	Spencer David Southern	

206648
Spencer Southern
333 E. Bristol Rd.
Delton, MI 49046

CONFIRMATION NO. 2156

FORMALITIES LETTER



OC000000088120632

Date Mailed: 05/06/2025

NOTICE TO FILE MISSING PARTS OF PROVISIONAL APPLICATION

FILED UNDER 37 CFR 1.53(c)

Filing Date Granted

An application number and filing date have been accorded to the above-identified provisional application. The items indicated below, however, are missing. Applicant is given **TWO MONTHS** from the date of this Notice within which to file all required items and pay any fees required below to avoid abandonment. Extensions of time may be obtained by filing a petition accompanied by the extension fee under the provisions of 37 CFR 1.136(a).

- The statutory basic filing fee is insufficient based on the current entity status of the application. The remainder of the basic filing fee or a proper establishment of a discounted entity status must be submitted.
- Surcharge as set forth in 37 CFR 1.16(g) has not been received.

The surcharge is due for late submission of filing fee or cover sheet.

SUMMARY OF FEES DUE:

The fee(s) required (including the fees noted above) are as follows. Small entity discount is in effect. If applicant is qualified for micro entity status, an acceptable Certification of Micro Entity Status must be submitted to establish micro entity status. (See 37 CFR 1.29 and forms PTO/SB/15A and 15B.)

- **\$130** basic filing fee.
- **\$26** surcharge.
- **\$(65)** previous unapplied payment amount.
- **\$91** TOTAL FEE BALANCE DUE.

Applicant is notified that the above-identified application contains the deficiencies noted below. No period for reply is set forth in this Notice for correction of these deficiencies. However, if a deficiency relates to the inventor's oath or declaration, the applicant must file an oath or declaration in compliance with 37 CFR 1.63, or a substitute statement in compliance with 37 CFR 1.64, executed by or with respect to each actual inventor no later than the expiration of the time period set in the "Notice of Allowability" to avoid abandonment. See 37 CFR 1.53(f).

Items Required To Avoid Processing Delays:

The deficiencies indicated below should be addressed promptly to avoid further processing delays.

This application, which was filed with an indication of micro entity status, fails to meet the requirements for establishing micro entity status because:

- Certification of micro entity status is missing. (See 37 CFR 1.29 and forms PTO/SB/15A and 15B.)

If micro entity status is established in reply to this notice by filing an acceptable Certification of Micro Entity Status, the fees itemized in the SUMMARY OF FEES DUE section of this Notice will be reduced to reflect the appropriate micro entity discount, and any previous payment amount may be applied to the discounted fees.

If applicant is not entitled to micro entity status under 37 CFR 1.29, applicant may still qualify for an appropriate reduction of fees if small entity status can be established under 37 CFR 1.27.

Replies must be received in the USPTO within the set time period or must include a proper Certificate of Mailing or Transmission under 37 CFR 1.8 with a mailing or transmission date within the set time period. For more information and a suggested format, see Form PTO/SB/92 and MPEP 512.

Replies should be addressed to:

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REPLY VIA ELECTRONIC FILING

A reply may be submitted via electronic filing only by registered users of the appropriate USPTO patent electronic filing system and must include:

- A separate document having the document description, "**Applicant response to Pre-Exam Formalities Notice**"; and
- the item(s) for which a reply period is set forth above.

For more information about electronic filing, contact the Patent Electronic Business Center (EBC) 1-866-217-9197 or visit <https://www.uspto.gov/ebc>

NOTE: If the reply is not being submitted electronically, a copy of this notice must be included.

Questions about the contents of this notice and the requirements it sets forth should be directed to the Office of Data Management, Application Assistance Unit, at **(571) 272-4000** or **(571) 272-4200** or **1-888-786-0101**.

/laguirre/

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- www.uspto.gov/ProSePatents

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- We hope you connect with us by subscribing to receive important notifications by email.
 - www.uspto.gov/subscribe



UNITED STATES
PATENT AND TRADEMARK OFFICE ®

TO: spencersouthern12@gmail.com
FROM: noreply@uspto.gov
CC: patentcenter_eofficeaction@uspto.gov
SUBJECT: USPTO: Patent Electronic System - Correspondence Notification for Customer Number 206648

Tue May 06 05:01:08 EDT 2025

Dear Patent Center Customer:

Correspondence Address:

Spencer Southern
333 E. Bristol Rd.
Delton, MICHIGAN, 49046
UNITED STATES

This is a courtesy notification regarding the following USPTO patent application(s) associated with your Customer Number, 206648, that have new outgoing correspondence. This correspondence is now available for viewing in Patent Center.

The official date of notification of the outgoing correspondence will be indicated on the form (e.g., PTOL-90) accompanying the correspondence.

Disclaimer:

The list of documents shown below are provided as a courtesy and is not part of the official file wrapper. The content of the images shown in the Image File Wrapper is the official record.

Application	Document	Mailroom Date	Attorney Docket No.
63788821	APP.FILE.REC	05/06/2025	
63788821	NTC.MISS.PRT	05/06/2025	
63788821	WELCOME.LET	05/06/2025	

To view your correspondence online, please sign in to [Patent Center](#) and then select Workbench/View correspondence. To update your email address(es), select Manage/Manage customer numbers.

If you have any questions, please contact the [Patent Electronic Business Center](#) (EBC) at ebc@uspto.gov or 866-217-9197 Monday – Friday, 6 a.m. to midnight ET.

Please do not reply to this email as it was sent from an unmonitored mailbox.

Sincerely,

The Patent Center Team



UNITED STATES
PATENT AND TRADEMARK OFFICE

P.O. Box 1450
Alexandria, VA 22313 - 1450
www.uspto.gov

ELECTRONIC ACKNOWLEDGEMENT RECEIPT

APPLICATION #
63/788,821

RECEIPT DATE / TIME
04/29/2025 03:40:17 PM Z ET

ATTORNEY DOCKET #
-

Title of Invention

Mint-to Logic System for Validation-Based Digital Routing and Protocol Execution

Application Information

APPLICATION TYPE	Utility - Provisional Application under 35 USC 111(b)	PATENT #	-
CONFIRMATION #	2156	FILED BY	SPENCER SOUTHERN
PATENT CENTER #	70215510	FILING DATE	-
CUSTOMER #	-	FIRST NAMED INVENTOR	Spencer David Southern
CORRESPONDENCE ADDRESS	Spencer David Southern, - 333 E. Bristol Rd., - Delton, MICHIGAN 49046 UNITED STATES	AUTHORIZED BY	-

Documents

TOTAL DOCUMENTS: 1

DOCUMENT	PAGES	DESCRIPTION	SIZE (KB)
correspondence.pdf	2	Change of Address	380 KB

Digest

DOCUMENT	MESSAGE DIGEST(SHA-512)
correspondence.pdf	E9D3B1B57765441AC9E77E17FB2008BBBAA2DC70940C0F94 2AFC0262B94D93E821E13034B1A922EB3AF42D88C035E225F 6858A35CBBB2996C9DA86A40C1AC43A

This Acknowledgement Receipt evidences receipt on the noted date by the USPTO of the indicated documents, characterized by the applicant, and including page counts, where applicable. It serves as evidence of receipt similar to a Post Card, as described in MPEP 503.

New Applications Under 35 U.S.C. 111

If a new application is being filed and the application includes the necessary components for filing date (see 37 CFR 1.53(b)-(d) and MPEP 506), a Filing Receipt (37 CFR 1.54) will be issued in due course and the date shown on this Acknowledgement Receipt will establish the filing date of the application

National Stage of an International Application under 35 U.S.C. 371

If a timely submission to enter the national stage of an international application is compliant with the conditions of 35 U.S.C. 371 and other applicable requirements a Form PCT/DO/EO/903 indicating acceptance of the application as a national stage submission under 35 U.S.C. 371 will be issued in addition to the Filing Receipt, in due course.

New International Application Filed with the USPTO as a Receiving Office

If a new international application is being filed and the international application includes the necessary components for an international filing date (see PCT Article 11 and MPEP 1810), a Notification of the International Application Number and of the International Filing Date (Form PCT/RO/105) will be issued in due course, subject to prescriptions concerning national security, and the date shown on this Acknowledgement Receipt will establish the international filing date of the application.

CHANGE OF CORRESPONDENCE ADDRESS <i>Application</i> Address to: Commissioner for Patents P.O. Box 1450 Alexandria, VA 22313-1450	Application Number	63/788,821
	Filing Date	04/15/2025
	First Named Inventor	Spencer David Southern
	Art Unit	
	Examiner Name	
	Attorney Docket Number	

Please change the Correspondence Address for the above-identified patent application to:



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I am the:



Applicant



Attorney or agent of record. Registration Number _____.



Registered practitioner named in the application transmittal papers who acts in a representative capacity under 37 CFR 1.34. See 37 CFR 1.33(a)(1).

Registration Number _____.

Signature

/ Spencer David Southern /

Typed or Printed
Name

Spencer David Southern

Date 04/29/2025

Telephone

2693318435

NOTE: This form must be signed in accordance with 37 CFR 1.33. See 37 CFR 1.4(d) for signature requirements and certifications. Submit multiple forms if more than one signature is required, see below*.



*Total of _____ forms are submitted.

Privacy Act Statement

The Privacy Act of 1974 (P.L. 93-579) requires that you be given certain information in connection with your submission of the attached form related to a patent application or patent. The United States Patent and Trademark Office (USPTO) collects the information in this record under authority of 35 U.S.C. 2. The USPTO's system of records is used to manage all applicant and owner information including name, citizenship, residence, post office address, and other information with respect to inventors and their legal representatives pertaining to the applicant's/owner's activities in connection with the invention for which a patent is sought or has been granted. The applicable Privacy Act System of Records Notice for the information collected in this form is COMMERCE/PAT-TM-7 Patent Application Files, available in the Federal Register at 78 FR 19243 (March 29, 2013), <https://www.govinfo.gov/content/pkg/FR-2013-03-29/pdf/2013-07341.pdf>.

Routine uses of the information in this record may include disclosure to: 1) law enforcement, in the event that the system of records indicates a violation or potential violation of law; 2) a federal, state, local, or international agency, in response to its request; 3) a contractor of the USPTO having need for the information in order to perform a contract; 4) the Department of Justice for determination of whether the Freedom of Information Act (FOIA) requires disclosure of the record; 5) a Member of Congress submitting a request involving an individual to whom the record pertains, when the individual has requested the Member's assistance with respect to the subject matter of the record; 6) a court, magistrate, or administrative tribunal, in the course of presenting evidence, including disclosures to opposing counsel in the course of settlement negotiations; 7) the Administrator, General Services Administration (GSA), or their designee, during an inspection of records conducted by GSA under authority of 44 U.S.C. 2904 and 2906, in accordance with the GSA regulations and any other relevant (i.e., GSA or Commerce) directive, where such disclosure shall not be used to make determinations about individuals; 8) another federal agency for purposes of National Security review (35 U.S.C. 181) and for review pursuant to the Atomic Energy Act (42 U.S.C. 218(c)); 9) the Office of Personnel Management (OPM) for personnel research purposes; and 10) the Office of Management and Budget (OMB) for legislative coordination and clearance.

If you do not furnish the information requested on this form, the USPTO may not be able to process and/or examine your submission, which may result in termination of proceedings, abandonment of the application, and/or expiration of the patent.

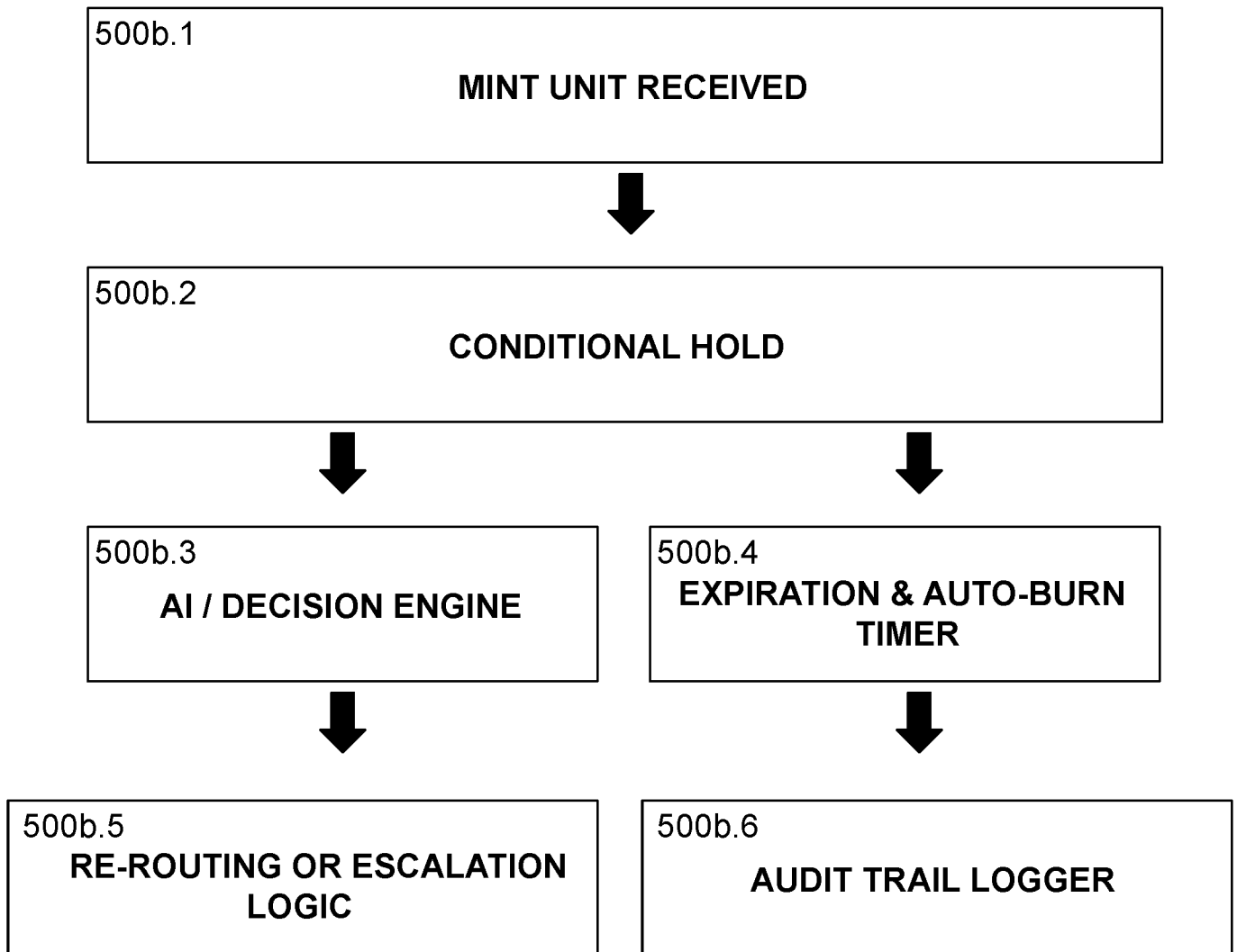
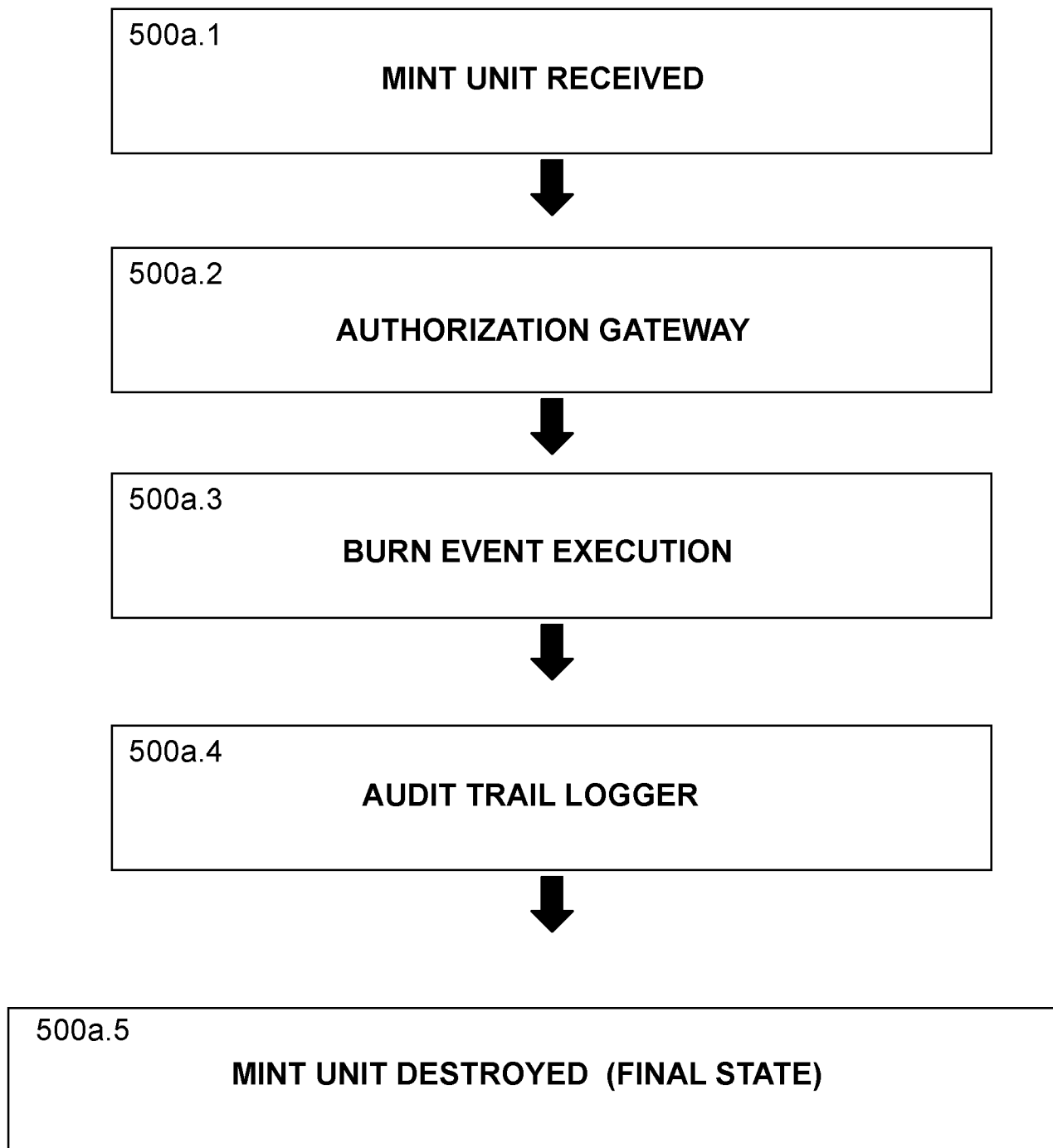


FIG. 7. EXPANDED VIEW OF MINT-TO-LIMBO PROTOCOL ARCHITECTURE
[500b]



**FIG. 6. EXPANDED VIEW OF MINT-TO-BURN PROTOCOL ARCHITECTURE
[500a]**

1. Title of the Invention

Mint-to Logic™ System for Validation-Based Digital Routing and Protocol Execution

2. Background of the Invention

Modern digital systems rely heavily on authentication and routing logic to determine user access, asset control, and protocol initiation. However, most current models lack modular validation frameworks, clean token lifecycle controls, and flexible routing strategies that can adapt to real-time contextual data or ethical AI systems.

Conventional blockchain and smart contract systems, while secure, often sacrifice adaptability, real-world triggers, and granular auditability. There exists a need for a validation-first, action-triggered system that mints unique execution tokens — known herein as Mint Units — only after confirmation, and routes them intelligently through defined protocol paths such as burn, limbo, access, or code execution.

This invention introduces a logic framework and system architecture called the Mint-to Logic system, which is designed to securely validate input, mint unique logic units, and route them through scalable, protocol-defined flows that can power everything from secure access to AI-driven tasks.

3. Summary of the Invention

The Mint-to Logic™ system is a modular protocol infrastructure that enables secure, validated digital routing through Mint Units™ — one-time logic objects that are created only after specific triggers are met. Once minted, these Mint Units are routed via embedded metadata to one of several defined execution protocols, such as burn (immediate destruction), limbo (conditional or delayed access), access (permission granting), or code (smart logic execution).

Each Mint Unit contains embedded routing instructions, timestamps, and validation metadata. The system is capable of interfacing with real-world actions, biometric inputs, environmental data, and AI systems to determine routing outcomes. This architecture provides a secure, scalable way to direct actions in both digital and physical systems — from entry tokens to AI commands — while preserving audit trails and maintaining control over each logic unit's lifecycle.

This system lays the foundation for next-generation identity management, programmable trust layers, event-based tokenization, and cross-protocol AI integration.

4. Brief Description of the Drawings

FIG. 1 is a high-level system architecture diagram of the Mint-to Logic framework, illustrating the sequential flow from source input through validation, Mint Unit generation, routing, protocol execution, and audit logging.

FIG. 2 is an expanded view of the Source Input architecture [100], depicting various input types including biometric, device-based, user data, environmental, and contextual sources.

FIG. 3 is an expanded view of the Validation / Trigger architecture [200], showing authentication layers, trigger mechanisms, and logic initiators leading to Mint Unit creation.

FIG. 4 is an expanded view of the Mint Unit Generation architecture [300], including unique identifier hashing, metadata encoding, timestamp assignment, and initial state control.

FIG. 5 is an expanded view of the Routing Decision Layer [400], illustrating the system logic that determines the protocol pathway the Mint Unit will follow based on embedded metadata and validation outcome.

FIG. 6 is an expanded view of the Mint-to-Burn Protocol architecture [500a], representing immediate-use, single-purpose tokens that are destroyed after a completed transaction or action.

FIG. 7 is an expanded view of the Mint-to-Limbo Protocol architecture [500b], illustrating conditional token holds, AI-evaluated decisions, time-based expiration, and rerouting pathways.

FIG. 8 is an expanded view of the Mint-to-Access Protocol architecture [500c], showing validation-based access control to physical, digital, or system assets, along with optional time/context restrictions and lifecycle finalization.

FIG. 9 is an expanded view of the Mint-to-Code Protocol architecture [500d], depicting logic engagement for smart contract triggers, AI instruction delivery, system automation, or external execution events.

5. Detailed Description of the Preferred Embodiments

[100] Source Input / Initiation

The system begins with a source input module [100], which serves as the primary intake point for any data or event intended to initiate a validation-triggered minting sequence. This input may originate from one or more sources including but not limited to user-submitted personal identifiers, biometric scans, device-based credentials, sensor data, time-sensitive triggers, financial information, or environmental conditions.

As illustrated in FIG. 2, the source input architecture is modular and designed to support multiple, parallel input types. For example, submodules [100.1], [100.2], and [100.3] may receive data from biometric readers, password or code entry fields, or device fingerprinting systems, respectively. Additional submodules [100.4] and [100.5] may include external API inputs, environmental signal receivers, or geolocation inputs. Optional component [100.6] may capture contextual meta-input such as time of day, network condition, or behavioral heuristics.

These submodules may function independently or in conjunction, forming a flexible, multi-channel intake layer that ensures the Mint-to Logic system is adaptable to evolving validation methods and scalable to numerous use cases. Each input type is recorded with a timestamp and routed to the validation subsystem [200] only when system readiness and intake completeness is confirmed.

[200] Validation / Trigger

The validation subsystem [200], as illustrated in FIG. 3, is responsible for evaluating incoming data from the source input layer [100] and determining whether the input satisfies pre-defined criteria necessary to initiate the minting of a Mint Unit™. This subsystem may incorporate single or multi-factor authentication mechanisms, such as biometric verification [200.1], passcode entry [200.2], or behavioral or environmental signal matching [200.3].

Component [200.4] represents optional AI-assisted validation, which may evaluate contextual data, risk profiles, or cross-reference real-time signals. Component [200.5] includes third-party verifiers or external validation modules which may confirm digital identity, payment authorization, or licensing credentials.

Once validation is successful, the system triggers Mint Unit creation via [200.6], the Mint Trigger Interface. This trigger acts as the transitional gate from passive input to active logic, ensuring that no Mint Unit is created without successful confirmation. Failed validations are rejected, logged, and optionally routed for further analysis or reattempt.

This layered, flexible structure allows for complex authentication stacks or simplified flows depending on application requirements.

[300] Mint Unit Generation

Once validation is confirmed, the Mint Unit Generation module [300], shown in FIG. 4, is activated. This component creates a unique Mint Unit—a digital logic object containing metadata used for downstream execution. Mint Units are single-use and may exist on-chain, off-chain, or within isolated system memory depending on application needs.

Subcomponent [300.1] generates a unique identifier (UID) or hash for the Mint Unit, while [300.2] timestamps its creation. Metadata encoding [300.3] embeds the results of the validation, including relevant user data, contextual parameters, and routing logic directives. Optionally, [300.4] applies state or expiration logic, limiting how long the Mint Unit™ is valid for execution.

Component [300.5] packages the Mint Unit in a secured wrapper, optionally cryptographically signed, to prevent tampering. The completed Mint Unit is then passed to the Routing Decision Layer [400] for directional logic evaluation.

[400] Routing Decision Layer

The Routing Decision Layer [400], depicted in FIG. 5, evaluates the metadata within the Mint Unit and determines which execution protocol it should follow. The embedded routing directive may point the Mint Unit to one of four primary logic paths: Mint-to-Burn [500a], Mint-to-Limbo [500b], Mint-to-Access [500c], or Mint-to-Code [500d].

Component [400.1] analyzes the Mint Unit's internal logic and validation context. It cross-references this with a rule engine [400.2], which may include static rules, AI-based logic trees, or user-defined flows. Component [400.3] generates the routing signal, which is embedded in the Mint Unit and used to direct it to the proper execution protocol. The routing decision is then logged for traceability and audit purposes.

This layer enables programmable, modular decision-making that adapts to different applications, such as secure facility access, NFT minting, AI logic deployment, or event ticketing.

[500a] Mint-to-Burn Protocol

The Mint-to-Burn Protocol [500a], illustrated in FIG. 6, is designed for use cases where the Mint Unit is intended to exist only briefly and is destroyed immediately after a single authorized use. This logic path is optimal for applications such as one-time access codes, single-use voting credentials, or expiring transaction keys.

Upon arrival at [500a.1], the Mint Unit is verified for structural integrity. Component [500a.2] confirms the protocol directive and ensures that it matches the routing instruction embedded at the routing layer [400.3]. Execution logic [500a.3] processes the Mint Unit as a one-time credential, initiating its function (e.g., granting access, logging vote, delivering content). Once execution is complete, the Mint Unit is passed to the burn operation [500a.4], which irreversibly invalidates or deletes it.

The final state [500a.5] logs this destruction in the audit ledger, ensuring full traceability without storing personal information (non-PII).

This protocol ensures maximum security, minimal resource retention, and strict one-time-use logic.

[500b] Mint-to-Limbo Protocol

The Mint-to-Limbo Protocol [500b], depicted in FIG. 7, provides a controlled state in which a Mint Unit is held temporarily under conditional access logic. This is useful for situations where AI must make a decision, additional user input is pending, or a time-based condition must be met.

When the Mint Unit enters [500b.1], it is assigned a limbo status and securely stored. Component [500b.2] monitors for conditional triggers such as a secondary verification, AI decision logic [500b.3], or time-delay [500b.4]. If no action is taken within a defined window, the system automatically invalidates the Mint Unit via [500b.5]. If conditions are satisfied, the Mint Unit is either rerouted to another protocol or allowed to proceed.

Optional return loops and recalibration pathways may be integrated here. All states, outcomes, and expiry events are recorded in [500b.6], the audit logger.

This protocol introduces flexibility, decision-making latency, and programmable pauses to enhance systemic integrity.

[500c] Mint-to-Access Protocol

The Mint-to-Access Protocol [500c], shown in FIG. 8, governs access to a physical, digital, or system-level asset upon successful validation and routing. It is commonly used in door locks, system login gates, content portals, or limited access zones.

Upon receiving a Mint Unit at [500c.1], the system runs a final access validator [500c.2] to ensure that conditions (e.g., time, location, identity) still align. Once cleared, access is granted via [500c.3]. Simultaneously, the system may enforce constraints such as time limits [500c.4] or resource scope, and it will initiate an access log through [500c.5].

After access is complete or time has expired, the Mint Unit is destroyed or archived through [500c.6].

This protocol guarantees secure, conditional entry access with immutable auditing.

[500d] Mint-to-Code Protocol

The Mint-to-Code Protocol [500d], as described in FIG. 9, enables the Mint Unit to act as an executable logic object, triggering internal or external code based on embedded instructions.

Once routed to [500d.1], the Mint Unit passes through a logic validator [500d.2] to confirm directive integrity and authorization. If approved, it enters the logic engagement engine [500d.3], which interprets and prepares the Mint Unit for execution. The command is passed to [500d.4], an execution module that may run system code, AI tasks, smart contracts, or webhook commands.

Results of this execution are captured by [500d.5] and logged for feedback, traceability, or future routing. The final state of the Mint Unit is handled in [500d.6], which finalizes or destroys it.

This protocol enables programmable trust, AI task control, and system automation from a secure tokenized entry point.

[600] Audit Trail and Logging

Component [600], referenced throughout the system, is responsible for recording every stage of the Mint Unit lifecycle in a secure, immutable manner. The audit trail preserves non-PII metadata including timestamps, session IDs, validation results, routing decisions, execution outcomes, and destruction confirmations.

Each Mint Unit event generates a corresponding log entry via [600.1], which includes a hash or UID, lifecycle state, protocol path taken, and any conditional metadata. Logs are stored in an encrypted or blockchain-compatible ledger via [600.2] and may be accessed for auditing, security review, or machine learning optimization.

The audit trail subsystem may also include alert triggers [600.3] for abnormal behavior, such as repeated access attempts, protocol re-entry, or tampering signals. The system ensures transparency without compromising user privacy, and it forms the accountability backbone of the Mint-to Logic infrastructure.

This module is optional but strongly recommended for regulated applications such as identity systems, digital voting, financial access controls, or AI transparency layers.

6. Informal Claims

1. A modular logic-based digital execution system, herein referred to as the Mint-to Logic™ system, comprising:

- (a) a validation subsystem configured to receive one or more data inputs from source input channels;
- (b) a trigger mechanism that activates upon successful validation;
- (c) a minting engine that creates a single-use logic object referred to as a Mint Unit;
- (d) a routing layer configured to evaluate the metadata within the Mint Unit and direct it to one or more protocol paths;

(e) at least one protocol module selected from a group consisting of: Mint-to-Burn, Mint-to-Limbo, Mint-to-Access, and Mint-to-Code;

(f) an audit subsystem for storing state, execution, and lifecycle metadata in an immutable or encrypted format.

2. The system of claim 1, wherein the validation subsystem supports biometric, environmental, contextual, password, device, or third-party verification methods.

3. The system of claim 1, wherein the Mint Unit includes a unique identifier, timestamp, validation metadata, and embedded routing logic.

4. The system of claim 1, wherein the Mint-to-Code protocol executes logic instructions embedded in or referenced by the Mint Unit, triggering smart contracts, system automation, or AI task execution.

5. The system of claim 1, wherein the Mint-to-Limbo protocol temporarily stores the Mint Unit in a conditional hold state pending additional input, AI analysis, or time-based events.

6. The system of claim 1, wherein the Mint-to-Burn protocol destroys the Mint Unit upon a single successful use and records the destruction in the audit trail.

7. The system of claim 1, wherein the Mint-to-Access protocol validates permission and grants conditional or time-based access to a secure resource or system state.

8. The system of claim 1, wherein each protocol path logs the execution result in the audit subsystem with non-personally identifiable metadata, session identifiers, and lifecycle state outcomes.

9. The system of claim 1, wherein Mint Units are generated on-chain, off-chain, or within isolated system environments depending on application context.

10. The system of claim 1, wherein the routing layer is dynamic and uses rule-based, AI-driven, or behaviorally adaptive logic to determine execution protocol.

7. Inventor Declaration and IP Disclaimer

I, Spencer Southern, am the sole inventor and developer of the system, components, and logic structures described in this application, including all figures, architectural frameworks, terminology, and associated use cases. I retain 100% ownership of all intellectual property related to this invention and its current or future modular applications, as outlined or referenced herein.

This document represents an original work under U.S. patent law and is being filed as a provisional patent application to secure priority rights.

All content herein is original intellectual property under development by the inventor. Unauthorized reproduction, use, or dissemination is prohibited. Patent(s) pending.

This system, known as the Mint-to Logic framework and including but not limited to the Mint Unit, associated protocols, and validation-to-routing architecture, is protected under provisional status from the date of official filing.

**Spencer Southern
Founder, Southern Star Pro. Studios, L.L.C.
Delton, Michigan
Date: 4/15/2025**

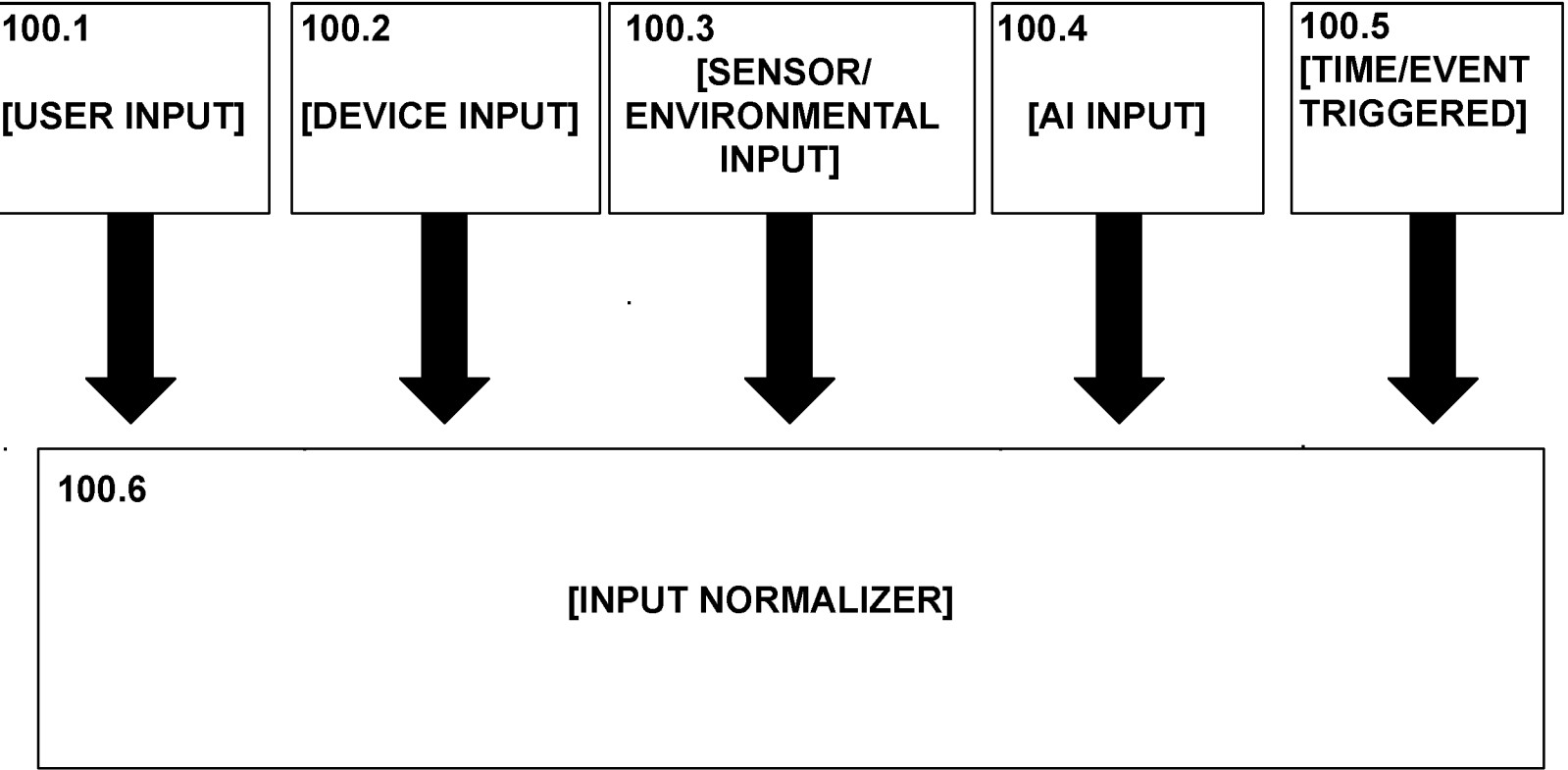


FIG. 2. EXPANDED VIEW OF SOURCE INPUT ARCHITECTURE [100]

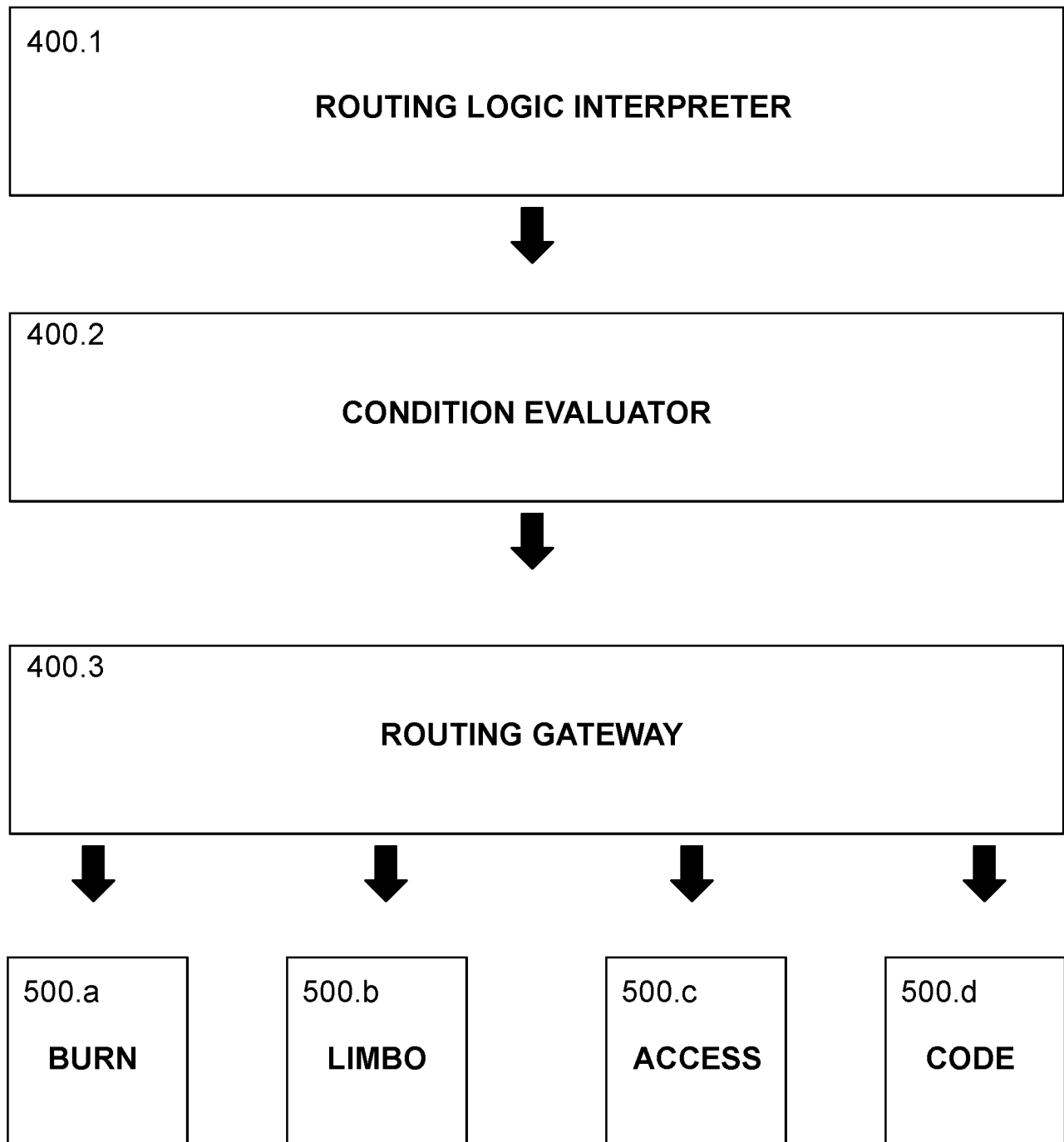
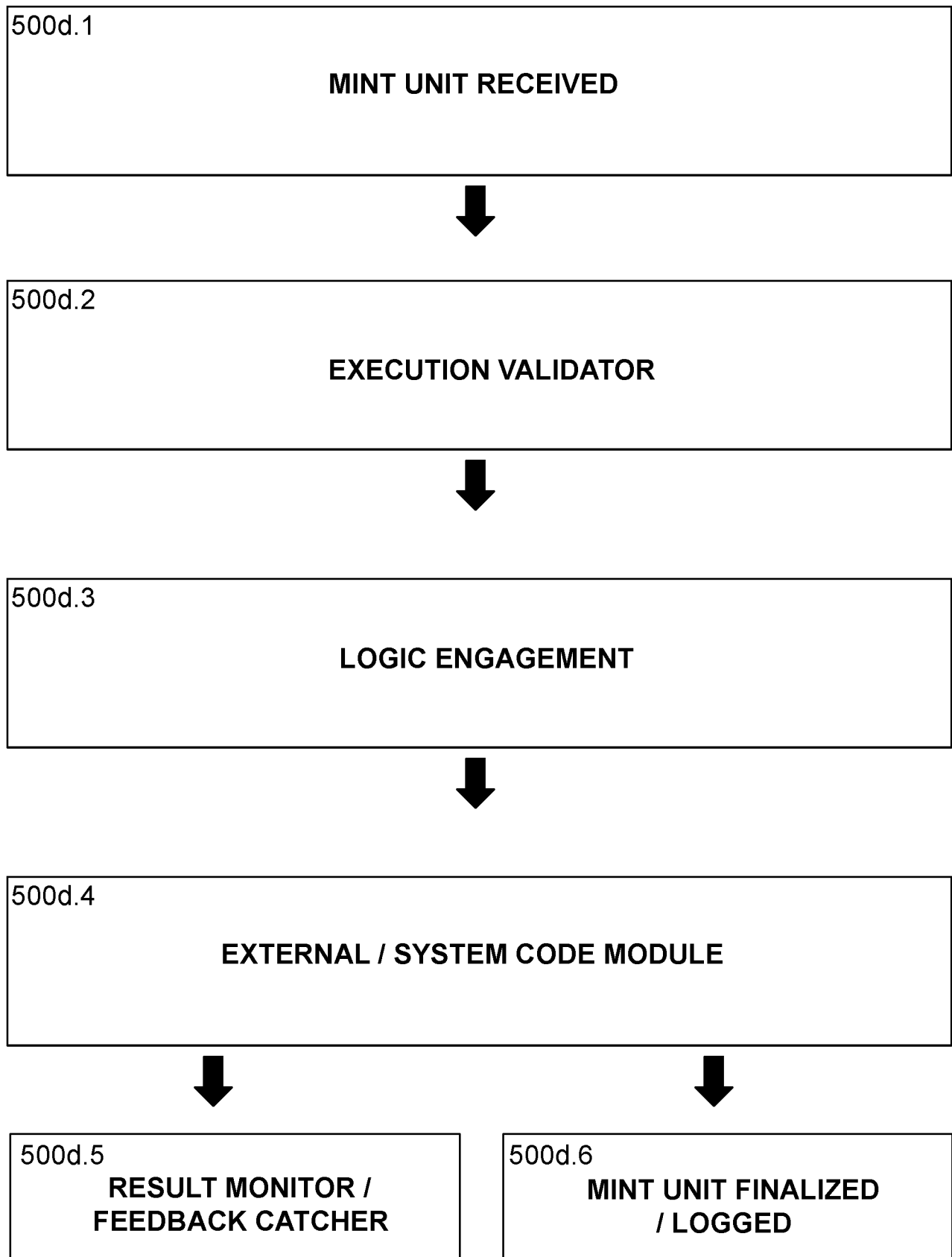
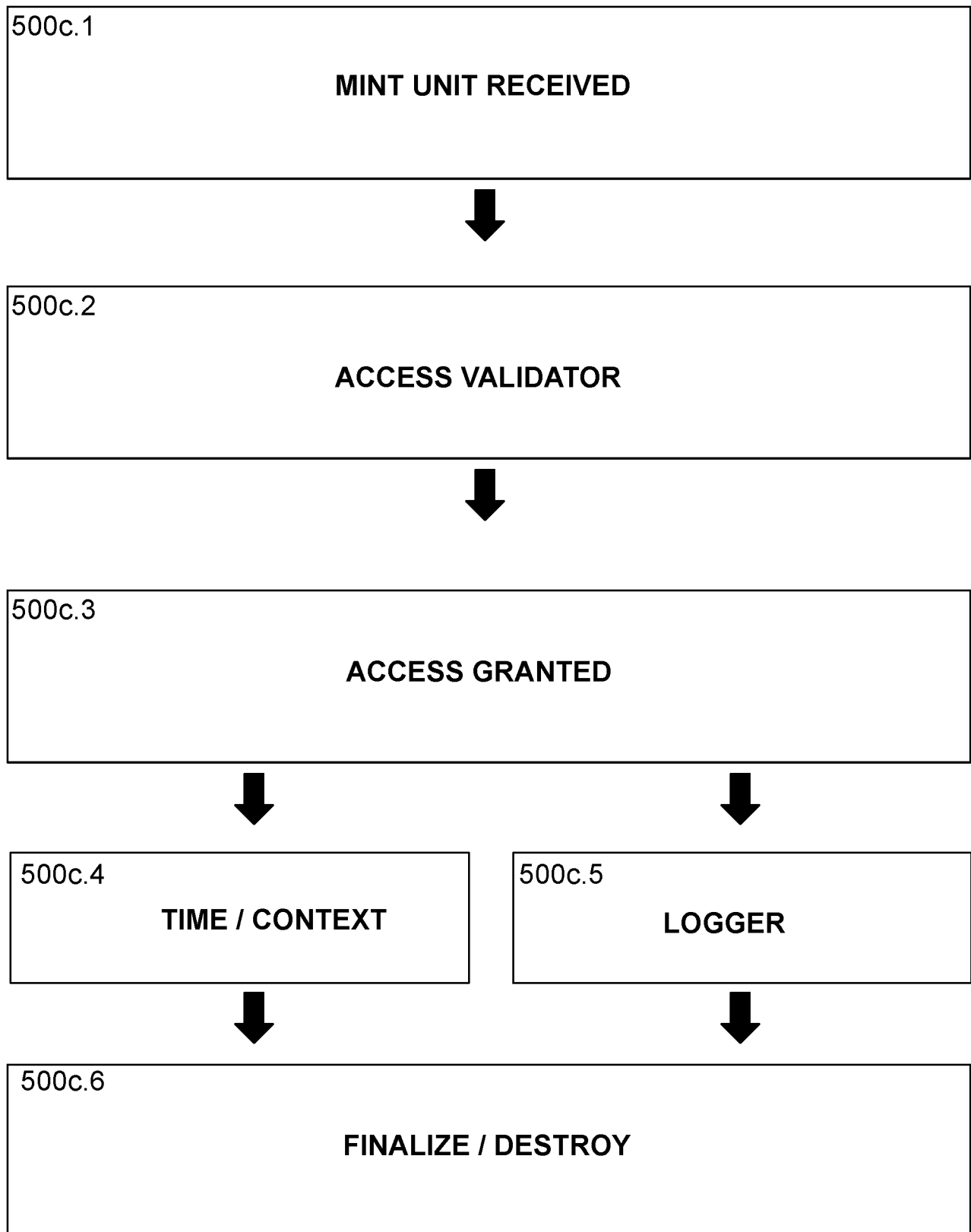


FIG. 5. EXPANDED VIEW OF ROUTING DECISION LOGIC [400]



**FIG. 9. EXPANDED VIEW OF MINT-TO-CODE PROTOCOL ARCHITECTURE
[500d]**



**FIG. 8. EXPANDED VIEW OF MINT-TO-ACCESS PROTOCOL ARCHITECTURE
[500c]**

Application Data Sheet 37 CFR 1.76

The Application Data Sheet is part of the provisional or non-provisional application for which it is being submitted. The following form contains the bibliographic data arranged in a format specified by the United States Patent and Trademark Office as outlined in 37 CFR 1.76.

Inventor Information

of inventors: 1

1. Spencer David Southern

Residence Information

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Application Information

Customer number ---

Correspondence address Spencer David Southern
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UNITED STATES

Title of invention **Mint-to Logic System for Validation-Based Digital Routing and Protocol Execution**

Attorney docket number ---

Small

Entity status

Application type Provisional Application under 35 USC 111(b)

Subject matter Utility

Total number of drawing sheets 9

Suggested figure for publication FIG. 1

Filing by reference No

Representative Information

of representatives: 0

Representative information should be provided for all practitioners having a power of attorney in the application. Providing this information in the Application Data Sheet does not constitute a power of attorney in the application (see 37 CFR 1.32).

 Data was not provided for this section.

Domestic Benefit/National Stage Information

of benefit claims: 0

This section allows for the applicant to either claim benefit under 35 U.S.C. 119(e), 120, 121, or 365(c), 386(c), or indicate National Stage entry from a PCT application. Providing benefit claim information in the Application Data Sheet constitutes the specific reference required by 35 U.S.C. 119(e) or 120, and 37 CFR 1.78(a)(2) or CFR 1.78(a)(4), and need not otherwise be made part of the specification.

 Data was not provided for this section.

Foreign Priority Information

of foreign priority claims: 0

This section allows for the applicant to claim benefit of foreign priority and to identify any prior foreign application for which priority is not claimed. Providing this information in the Application Data Sheet constitutes the claim for priority as required by 35 U.S.C. 119 (b) and 37 CFR 1.55. When priority is claimed to a foreign application that is eligible for retrieval under the priority document exchange program (PDX) the information will be used by the Office to automatically attempt retrieval pursuant to 37 CFR 1.55(i)(1) and (2). Under the PDX program, applicant bears the ultimate responsibility for ensuring that a copy of the foreign application is received by the Office from the participating foreign intellectual property office, or a certified copy of the foreign priority application is filed, within the time period specified in 37 CFR 1.55(g)(1).

 Data was not provided for this section.

Statement under 37 CFR 1.55 or 1.78 for AIA (First Inventor to File) Transition Applications

Checking this box will cause the application to be examined under the first inventor to file provisions of the AIA.

☐

This application (1) claims priority to or the benefit of an application filed before March 16, 2013 and (2) also contains, or contained at any time, a claim to a claimed invention that has an effective filing date on or after March 16, 2013.

NOTE: By providing this statement under 37 CFR 1.55 or 1.78, this application, with a filing date on or after March 2016, 2013, will be examined under the first inventor to file provisions of the AIA.

Authorization or Opt-Out of Authorization to Permit Access

When this Application Data Sheet is properly signed and filled with the application, applicant has provided written authority to permit a participating foreign intellectual property (IP) office access to the instant application-as-filed (see paragraph A in subsection 1 below) and the European Patent Office (EPO) access to any search results from the instant application (see paragraph B in subsection 1 below).

Should applicant choose not to provide an authorization identified in subsection 1 below, applicant **must opt-out** of the authorization by checking the corresponding box A or B or both in subsection 2 below.

NOTE:

This section of the Application Data Sheet is **ONLY** reviewed and processed with the **INITIAL** filing of an application. After the initial filing of an application, an Application Data Sheet cannot be used to provide or rescind authorization for access by a foreign IP office(s). Instead, Form PTO/SB/39 or PTO/SB/69 must be used as appropriate.

1. Authorization to Permit Access by a Foreign Intellectual Property Office(s)

Priority Document Exchange (PDX)

- A. Unless box A in subsection 2 (opt-out of authorization) is checked, the undersigned hereby **grants the USPTO authority** to provide the European Patent Office (EPO), the Japan Patent Office (JPO), the Korean Intellectual

Property Office (KIPO), the China National Intellectual Property Administration (CNIPA), the World Intellectual Property Organization (WIPO), and any other foreign intellectual property office participating with the USPTO in a bilateral or multilateral priority document exchange agreement in which a foreign application claiming priority to the instant patent application is filed, access to: (1) the instant patent application-as-filed and its related bibliographic data, (2) any foreign or domestic application to which priority or benefit is claimed by the instant application and its related bibliographic data, and (3) the date of filing of this Authorization. See 37 CFR 1.14(h) (1).

Search Results from U.S. Application to EPO

Unless box B in subsection 2 (opt-out of authorization) is checked, the undersigned hereby **grants the USPTO authority** to provide the EPO access to the bibliographic data and search results from the instant patent application when a European patent application claiming priority to the instant patent application is filed. See 37 CFR 1.14(h)(2).

B.

The applicant is reminded that the EPO's Rule 141(1) EPC (European Patent Convention) requires applicants to submit a copy of search results from the instant application without delay in a European patent application that claims priority to the instant application.

2. Opt-Out of Authorizations to Permit Access by a Foreign Intellectual Property Office(s)

☐

A. Applicant **DOES NOT** authorize the USPTO to permit a participating foreign IP office access to the instant application-as-filed. If this box is checked, the USPTO will not be providing a participating foreign IP office with any documents and information identified in subsection 1A above.

☒

B. Applicant **DOES NOT** authorize the USPTO to transmit to the EPO any search results from the instant patent application. If this box is checked, the USPTO will not be providing the EPO with the search results from the instant application.

NOTE:

Once the application has published or is otherwise publicly available, the USPTO may provide access to the application in accordance with 37 CFR 1.14.

Applicant Information

of applicants: 1

The information to be provided in this section is the name and address of the legal representative who is the applicant under 37 CFR 1.43; or the name and address of the assignee, person to whom the inventor is under an obligation to assign the invention, or person who otherwise shows sufficient proprietary interest in the matter who is the applicant under 37 CFR 1.46.

1. **Spencer David Southern**

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Applicant Individual
Applicant type Assignee

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Fax: ---

Email:

spencersouthern12@gmail.com

Assignee Information including Non-Applicant Assignee Information

of assignees: 1

An assignee-applicant identified in the "Applicant" section will appear on the patent application as an applicant.

Providing assignment information in this section does not substitute for compliance with any requirement of part 3 of Title 37 of CFR to have an assignment recorded by the Office.

Spencer David Southern

333 E. Bristol Rd.

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12693318435





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Individual

Signature

NOTE:

This Application Data Sheet must be signed in accordance with 37 CFR 1.33(b). **However, if this Application Data Sheet is submitted with the INITIAL filing of the application and either box A or B is not checked in subsection 2 of the "Authorization or Opt-Out of Authorization to Permit Access" section, then this form must also be signed in accordance with 37 CFR 1.14(c)**

This Application Data Sheet **must** be signed by a patent practitioner if one or more of the applicants is a **juristic entity** (e.g., corporation or association). If the applicant is two or more joint inventors, this form must be signed by a patent practitioner, **all** joint inventors who are the applicant, or one or more joint inventor-applicants who have been given power of attorney (e.g., see USPTO Form PTO/AIA/81) on behalf of **all** joint inventor-applicants.

See CFR 1.4(d) for the manner of making signatures and certifications.

Signature	First name	Last name	Registration #	Date
/Spencer Southern/	Spencer	Southern	---	04/15/2025

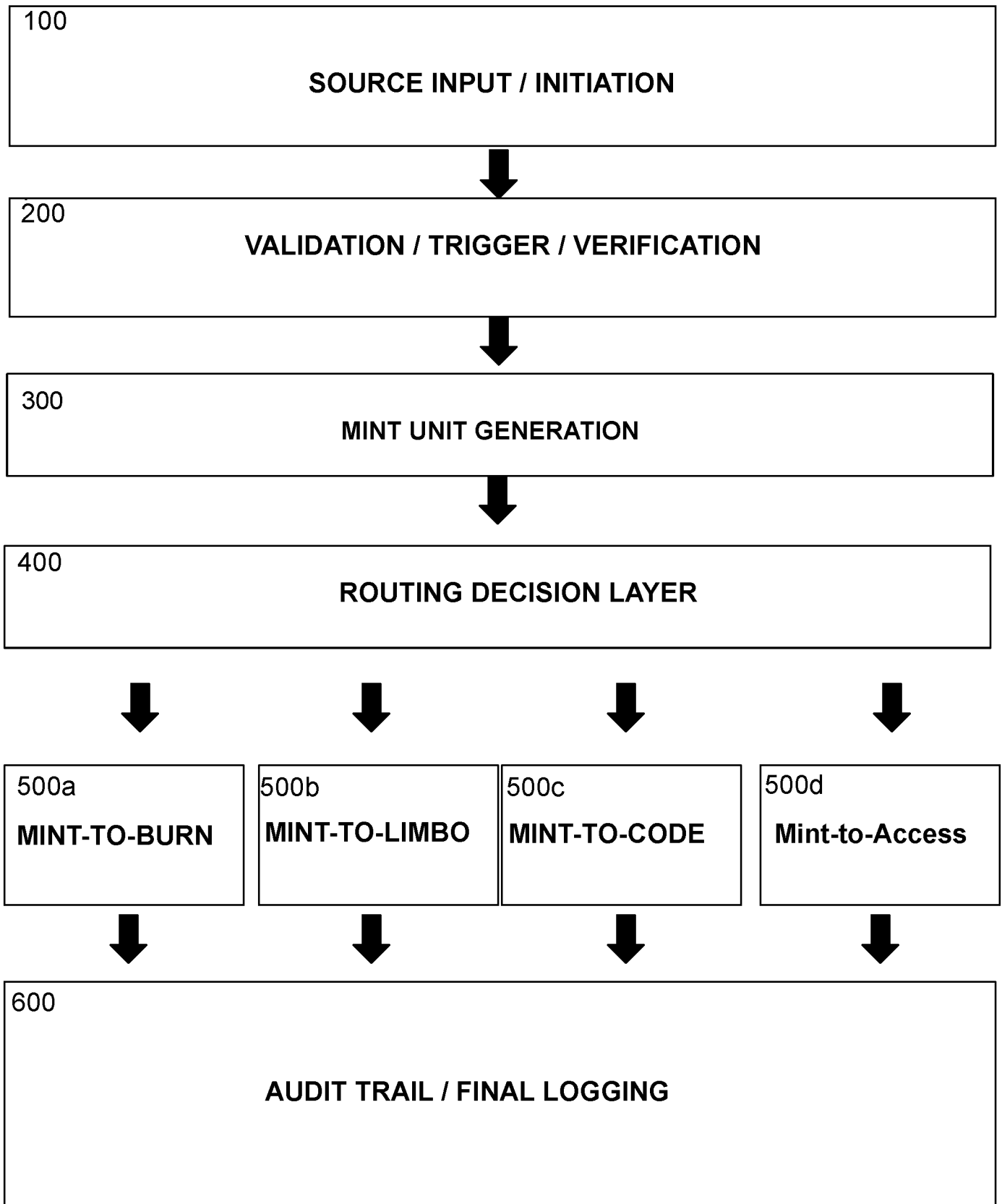


FIG. 1. MINT UNIT LIFECYCLE SYSTEM ARCHITECTURE FLOW DIAGRAM

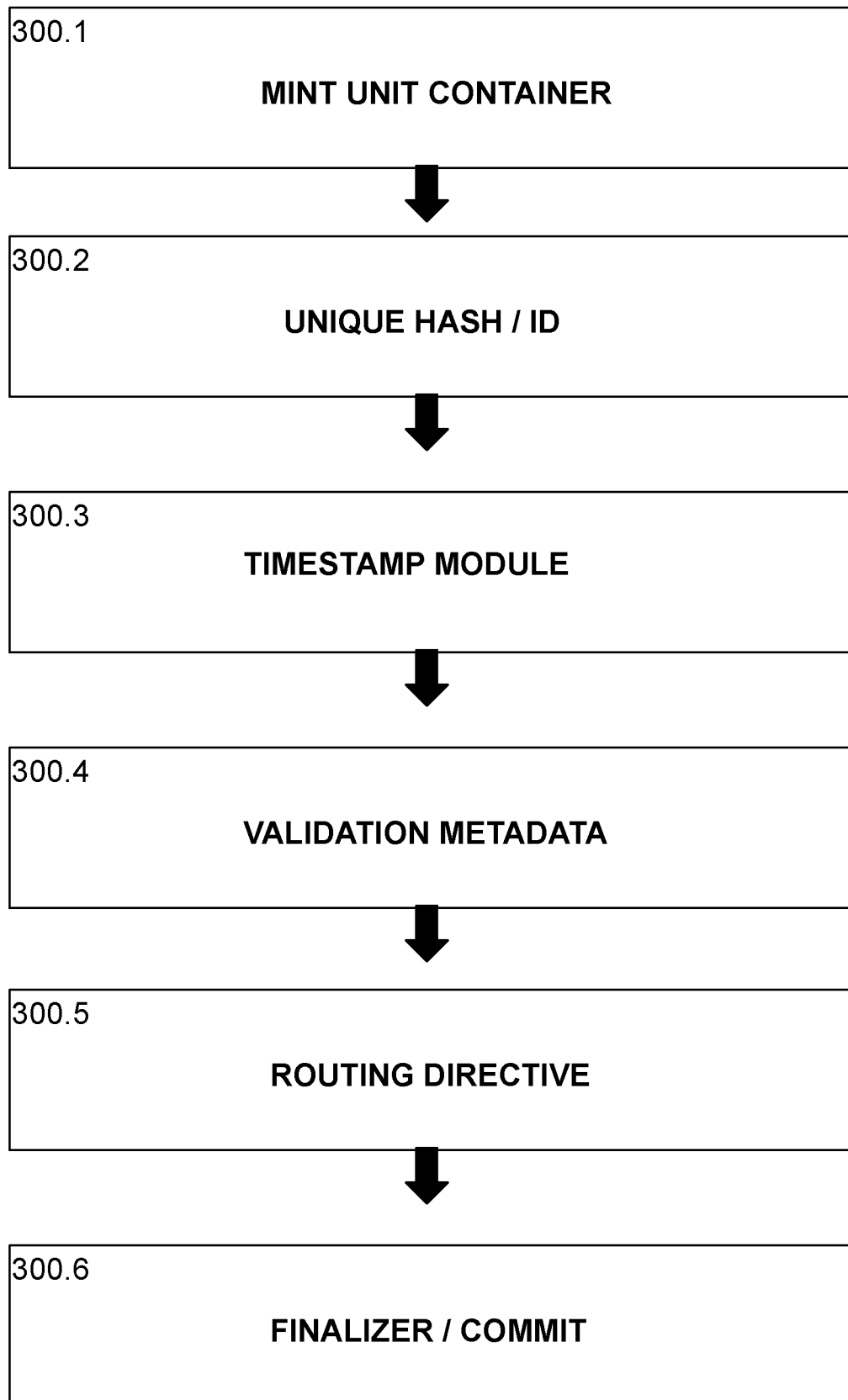


FIG. 4. EXPANDED VIEW OF MINT UNIT GENERATION ARCHITECTURE [300]

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ELECTRONIC ACKNOWLEDGEMENT RECEIPT

APPLICATION #
63/788,821RECEIPT DATE / TIME
04/15/2025 02:43:51 AM Z ETATTORNEY DOCKET #
-

Title of Invention

Mint-to Logic System for Validation-Based Digital Routing and Protocol Execution

Application Information

APPLICATION TYPE Utility - Provisional Application under
35 USC 111(b)

PATENT # -

CONFIRMATION # 2156

FILED BY Spencer Southern

PATENT CENTER # 70004449

FILING DATE -

CUSTOMER # -

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Documents

TOTAL DOCUMENTS: 11

DOCUMENT	PAGES	DESCRIPTION	SIZE (KB)
generatedADS70004449.pdf	5	Application Data Sheet	113 KB
Fig. 1 Expanded View.pdf	1	Drawings-only black and white line drawings	38 KB
FIG. 2. EXPANDED VIEW.pdf	1	Drawings-only black and white line drawings	32 KB
FIG. 3. EXPANDED VIEW.pdf	1	Drawings-only black and white line drawings	34 KB
FIG. 4 EXPANDE VIEW.pdf	1	Drawings-only black and white line drawings	34 KB
FIG. 8 EXPANDED VIEW.pdf	1	Drawings-only black and white line drawings	35 KB

FIG. 5 EXPANDED VIEW.pdf	1	Drawings-only black and white line drawings	35 KB
FIG. 6 expanded view.pdf	1	Drawings-only black and white line drawings	35 KB
FIG. 9 expanded view.pdf	1	Drawings-only black and white line drawings	36 KB
Mint-to Logic System for Validation-Based Digital Routing and Protocol Execution.pdf	8	Specification	120 KB
FIG. 7 EXPANDED VIEW.pdf	1	Drawings-only black and white line drawings	35 KB

Digest

DOCUMENT

MESSAGE DIGEST(SHA-512)

generatedADS70004449.pdf	6ED0A5C428044C397D8ECC3961998EDBF12701C815B75F28 C66CCB21480BD20439B3E599DCA0FAFB6671CC7804463C1E 50F17B802F41367FB5FD1DD88F558EB6
Fig. 1 Expanded View.pdf	C9D6D2B519C84570EB357E3993A793F81B06AAEACAE88B59 8D6ECC6899C8590712C71A7EA12C7C50A8E19D8464CAF50E 25BADFAED01C544F030B23BC250600BA
FIG. 2. EXPANDED VIEW.pdf	B6B27B6013FD1E1F40625A1EAB96810265EC367B3D7A0CEB2 7FE9EFE7C251516E1107504E600AA2AA8D3456A732D11B73E F850D23DE39286AE508418C09F0F2B
FIG. 3. EXPANDED VIEW.pdf	3C61CFCE30A507CBA73AC965D40D6E135382A02360B341006 C952D96B09BF86B922D723FDD07588D7748FA1A77EEFE2425 B44036804C94EF033A96FB1A044C32
FIG. 4 EXPANDE VIEW.pdf	07CA91E4CA7E59DEC58B60AFE9B0A98572FF72B13A2354857 7DBD748F54AC27FD5AE685E2826C61D06066713F102F5C7C8 D4CDACA2EC400D82AFDA38ECA57B7B
FIG. 8 EXPANDED VIEW.pdf	303FDEC3460D7795D8AA27FB5342A49F504799B9E736BEF16 3E802FC211D08CDAB84012225223CAF47A892D7580199D64C 2C9377F1973BF8EBEA12816F68A26E
FIG. 5 EXPANDED VIEW.pdf	3329778D1E42928027755D6EA58DC6FA07BAC9C48C3866D11 7B9497E307692C2336624B9C95CFC249A646577BF63AF958D CA3C4BD4D67CAFF21E66BF7FDB63D5
FIG. 6 expanded view.pdf	B630281782F88768FC32FF8274B3D451D27A1DCF9A3860CC2 457767CA4639D711B412F17C178DB785B4039312DCD65C5B0

FFCE34CCBC6A4846586CCD6345F55B

FIG. 9 expanded view.pdf

C190BCC3788B8D67B853908CB5FD4B6FA582C01004776E8B
B883113111321D186BC09188A0ED62FDABBC3846AC41B63E8
013C3B4D016DA9AB66904F0E259C8DBMint-to Logic System for
Validation-Based Digital
Routing and Protocol
Execution.pdf40B3AE5824D6FC9FBED312AA03D6E68957B0A8598DDDAEC
ED18F8D75858ED8D99CF8D43AB853B924ED2358E117B1F395
22CB128AAC44F31F88249E131848F0B5

FIG. 7 EXPANDED VIEW.pdf

0CD5056630CE950DD43A9820DADE7A09B911DA4FC70700B1
54F2D5B49B546FC739657C155234F5ABB6D235FEFFA590849
D56FD30BCD3A44F713B706BA9B46808

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If a new application is being filed and the application includes the necessary components for filing date (see 37 CFR 1.53(b)-(d) and MPEP 506), a Filing Receipt (37 CFR 1.54) will be issued in due course and the date shown on this Acknowledgement Receipt will establish the filing date of the application

National Stage of an International Application under 35 U.S.C. 371

If a timely submission to enter the national stage of an international application is compliant with the conditions of 35 U.S.C. 371 and other applicable requirements a Form PCT/DO/EO/903 indicating acceptance of the application as a national stage submission under 35 U.S.C. 371 will be issued in addition to the Filing Receipt, in due course.

New International Application Filed with the USPTO as a Receiving Office

If a new international application is being filed and the international application includes the necessary components for an international filing date (see PCT Article 11 and MPEP 1810), a Notification of the International Application Number and of the International Filing Date (Form PCT/RO/105) will be issued in due course, subject to prescriptions concerning national security, and the date shown on this Acknowledgement Receipt will establish the international filing date of the application.

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ELECTRONIC PAYMENT RECEIPT

APPLICATION #
63/788,821RECEIPT DATE / TIME
04/15/2025 02:43:51 AM Z ETATTORNEY DOCKET #
-

Title of Invention

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INVENTOR

Payment Information

PAYMENT METHOD
CARD / 8346PAYMENT TRANSACTION ID
E20254E346138903PAYMENT AUTHORIZED BY
Spencer Southern

FEE CODE	DESCRIPTION	ITEM PRICE(\$)	QUANTITY	ITEM TOTAL(\$)
3005	PROVISIONAL APPLICATION FILING FEE	65.00	1	65.00
TOTAL AMOUNT:				\$65.00

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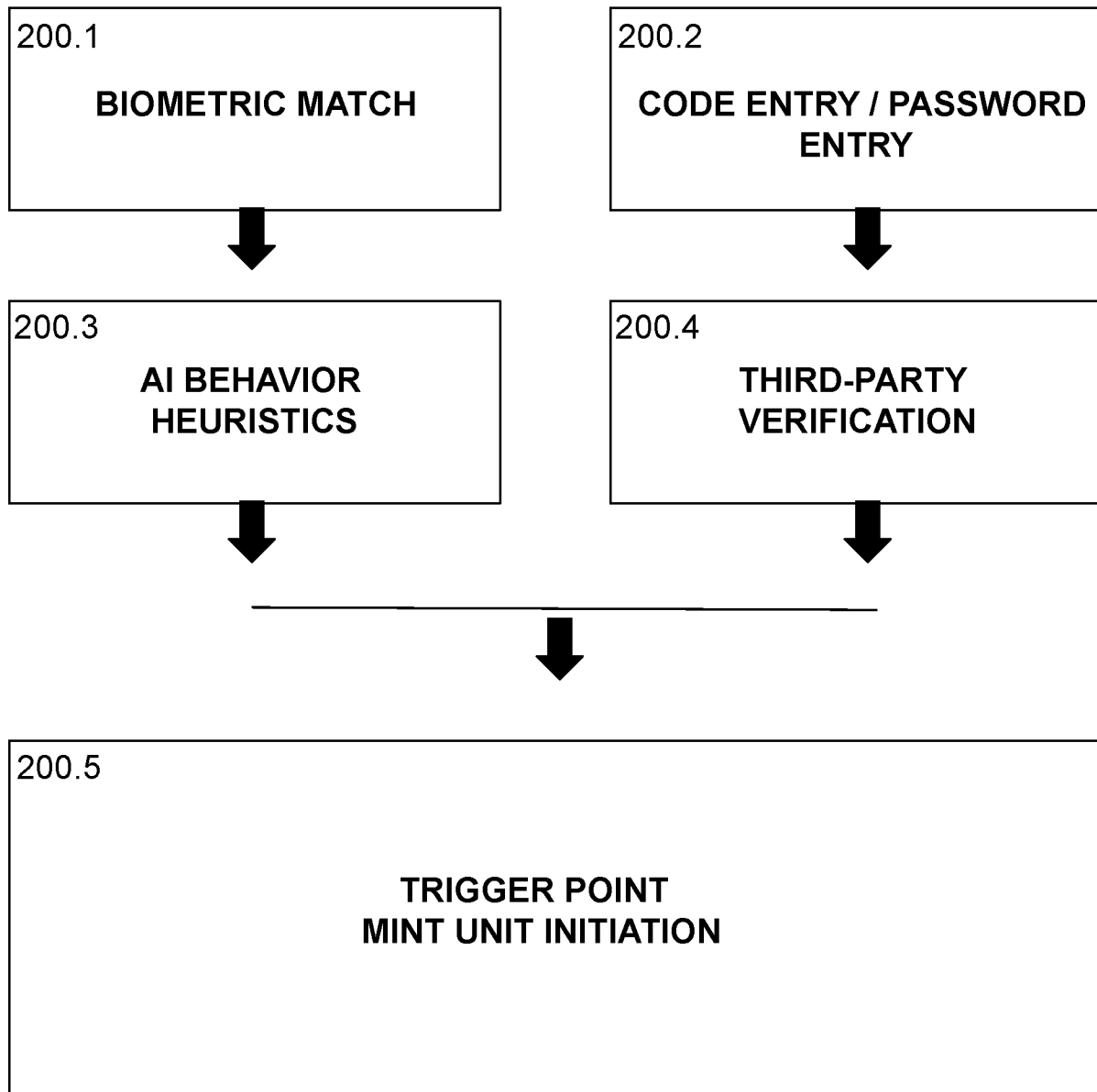


FIG. 3. EXPANDED VIEW OF VALIDATION / TRIGGER / VERIFICATION [200]

Placeholder Sheet for Supplemental Content

Application Number: 63788821

Document Date: 04/15/2025

The presence of this form indicates that the following document type was received in electronic format on the date identified above. This document is stored separately from the typical case contents which are stored in image format.

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